INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 4: Implementation and Empirical Results

by

TAEHONG KWON

Student number: 28576810

submitted in accordance with the requirements for the degree of

DOCTOR OF PHILOSOPHY

in the subject

COMPUTER SCIENCE

at the

UNIVERSITY OF SOUTH AFRICA

SUPERVISOR: Prof. Ernest Mnkandla

CO-SUPERVISOR: Prof. Donatien Koulla Moulla

CO-SUPERVISOR: Dr. David Sena Attipoe

October 2026

4 Implementation and Empirical Results

This chapter runs the design of Chapter 3 on Arbitrum One as a single case study: one chain and one observation window, one matched cohort of contracts for every same-window comparison, spender and trader cohorts for the two holdout windows W0 and W1, and subsamples of the matched cohort for runtime against size. It reports the revised analysis of Section 3.5.1, in which every amount is valued in US dollars and the cohort, the graphs, the proxies and the spender labels keep the five tokens WETH, USDC, USDT0, ARB and WBTC. The revision was decided after the registered results of the same window, W0 and W1 were known, so no result in this chapter is a registered result. Rule A was fixed in the analysis plan of 9 October 2026 before any log of the study was extracted; rule B and the comparison on outcomes built from neither edge type were registered after the W0 results of the registered analysis were known and before the study extracted the logs of W1. This chapter applies these rules, with their thresholds and wording, to the revised scores: rule A on W0 and in the same window, rule B on W1, and the comparison on W1 with W0 as its secondary window (Sections 4.6.2 and 4.6.3). Their verdicts here are a re-analysis, not registered tests; the registered analysis reached the same verdicts (Appendix 8.6.10, Table 8.13). Every table is generated from the evaluation summaries of Section 3.13, each Kendall τ comes with its 95% confidence interval except in the value-slope and sample-definition checks, which were computed without intervals, and each PageRank is read against the raw degree it smooths. Each section states which of its comparisons the plan or the registration named.

4.1 Case study setting: Arbitrum One

Arbitrum One was chosen because it is an Ethereum rollup on which many DeFi protocols run (Section 2.2), because its logs are in the public BigQuery export that the pipeline reads, and because

one of its margin-trading protocols, GMX V2, records the position closes behind the trader labels. The observation window runs from 1 October 2023 to 30 June 2026 (33 months), and the time decay is anchored at its end, T_{obs}. An event at the anchor has time weight 0.858; the weight falls to 0.5 at 180 days and is close to zero for events more than two years old (Figure 3.3). Events from the early months still create edges, but the last year carries most of the edge weight in AWP and EndorseRank.

4.1.1 Cohorts

Four sets of contracts are scored or evaluated. Every address in them holds contract code; externally owned accounts (EOAs) enter the graphs as owners, senders and recipients and are never scored.

- Matched cohort ($n = 15{,}107$), defined in Section 3.2: the contracts that received a non-zero `Approval` of one of the five tokens from at least three distinct owners. They are part of the plan's cohort of 27,844 contracts, which applied the same rule to the approvals of every token; its other 12,737 contracts do not meet the rule on the five tokens. Of the plan's 29,022 candidate spenders, `eth_getCode` at block 513,216,506 (9 October 2026) returned contract code for 27,844, no code for 1,121 and an EIP-7702 delegation designator (Buterin et al., 2024) for 57; the last two groups are EOAs and were left out. Membership does not depend on either graph at T_{obs}. Of the cohort contracts, 523 are outside the allowance graph, because by the end of the window the latest `Approval` of every owner and token that had approved them carried the value zero and they held no positive allowance of the five tokens as owners. The transfer graph leaves out 2,670, because no non-zero transfer of the five tokens in the window names them as sender or recipient (a spender that moves tokens with `transferFrom` from an owner directly to a third address does not appear in the resulting `Transfer` event), and 80 contracts are outside both graphs. A contract absent from a graph scores zero under that graph's method. Every same-window alignment, difference test and benchmark is computed on this cohort.
- Spender cohorts of the two holdout windows. In W0 the scores are frozen at t_1 (31 March 2026,

23:59:59 UTC) and the labels are counted from 1 April to 30 June 2026. The cohort is every contract spender that holds at least one positive latest allowance of one of the five tokens at t_1 in the extracted approval data. Of the 17,167 such spenders, 17,050 hold contract code, 104 have no code and 13 carry an EIP-7702 designator; the 117 EOAs are left out. In W1 the scores are frozen at T_{obs} (30 June 2026, 23:59:59 UTC), the labels are counted from 1 July to 30 September 2026, and the same rule gives 18,152 contract spenders. The spender cohorts are not subsets of the matched cohort. The extracted approvals include those that cohort contracts grant as owners, so the contracts they approve enter as spenders, and a spender needs only one owner with a positive allowance.

- Trader cohorts. The GMX V2 accounts that hold contract code, are nodes of the freeze-date graph (either layer) and close at least three positions in the label window: 88 in W0, 40 of which had received an approval by the freeze, and 70 in W1, 37 of which had. Contract accounts closed 133,685 GMX V2 positions in the observation window, 6,473 of them by liquidation, with 88,002 closes in 2025 against 9,809 in the first half of 2026, and 6,805 from July to September 2026, 182 of them by liquidation. Every trader result therefore rests on few contracts and has wide intervals.
- Subsamples of the matched cohort. Deterministic SHA256 subsamples (seed `contract-benchmark-v1`) with $n = 1{,}000$, $2{,}000$, $5{,}000$ and $10{,}000$ contracts, followed by the full cohort of 15,107 (Section 3.12.1). Each stage keeps the extracted edges incident to its sampled contracts, so its edge count grows with n, from 424,888 allowance and 1,692,513 transfer edges at $n = 1{,}000$ to the full graphs. The stages serve the runtime against size (Section 4.2.1) and the sensitivity to the sample definition (Section 4.5.3).

The five tokens hold most of the extracted rows. Of the ego events of the plan's cohort in the observation window, the revised analysis keeps 224,954,202 approvals (57.7%) and 1,149,168,538 transfers (74.7%). It leaves out 164,202,859 approvals and 364,468,917 transfers of other tokens, and 709,821 approvals and 24,766,612 transfers of the five tokens in which no contract of the revised

cohort takes part; every kept transfer has a price. At T_{obs}, 5,763,757 of the 12,590,613 positive latest allowances (45.8%) are unlimited, and 6,776,289 (53.8%) reach the cap U of the allowance layer of C-PR. On 38 sampled dates the DefiLlama prices differ from the Chainlink feeds read on the chain by a median of 0.01% to 0.08% per token and by at most 0.72%. Table 8.4 in Appendix 8.4.2 gives these figures per token, with the addresses and the feeds.

At T_{obs} the allowance graph has 6,221,665 nodes and 11,210,114 edges, and the transfer graph 7,884,602 nodes and 25,001,216 edges; at t_1 the two pair tables hold 10,650,258 allowance and 23,680,692 transfer pairs. The edge counts at T_{obs} are those of pairs with a positive EndorseRank or AWP weight. The node counts also include 2,602 and 2,246 addresses whose only pairs carry amounts so small that the value weight is zero in floating-point arithmetic; the solver graphs of EndorseRank and AWP leave them out (Table 4.1). On the same contracts the transfer graph thus has 2.23 times as many edges as the allowance graph, which has 44.8% of the transfer graph's edge count. An owner–spender pair is an allowance edge only while its latest allowance of one of the five tokens is positive, whereas every pair of addresses that exchanged a non-zero transfer of these tokens in the window is a transfer edge. Of the allowance graph's nodes, 6,208,769 are owners with at least one positive allowance, so that graph consists almost entirely of owners pointing at a much smaller set of spenders. Both graphs come from the same extraction, the ego network of the revised cohort, so the differences between the scores trace back to their edges and restarts.

4.1.2 Implementation

Every algorithm runs on one Python implementation of PageRank ($d = 0.85$, $\varepsilon = 10^{-8}$ in the L_1 norm, $I_{\max} = 300$). EndorseRank and AWP differ in their edge lists and restart vectors, and C-PR mixes its two edge lists node by node (Equation (3.9)). Each evaluation run writes its scores, per-proxy statistics and bootstrap intervals to one summary file, and the benchmark writes its timings to a file of its own. The registered analysis computed its same-window and W0 results after the observation window had been extracted and before W1 was registered, and its W1 results after the July–September logs had been extracted. Every result of the revised analysis was computed on

10 October 2026, from 11:36 UTC, after the revision had been committed and after the registered results of all three windows were known (Section 3.5.1). At $d = 0.85$ every solve converged well inside the iteration cap: at T_{obs} EndorseRank needed 69 iterations, AWP 92, C-PR at $\lambda = 0.5$ 79 and C-PR ($\lambda = 0$) 92, and the longest single solve, the seeded transfer walk of S-PR, 95 (Section 4.2). Appendix 8.1 lists the runs, the repository revision, the configuration file, the summary files and the machine on which the benchmark was timed.

4.2 Computational benchmark (Claim C1)

Claim C1 concerns what each score costs to compute on the same input. Following the timing procedure of Section 3.11, Table 4.1 reports for each algorithm, on the full graphs of the matched cohort at T_{obs}, the mean solve time and its standard deviation over five timed runs after one untimed warm-up, the peak memory of the warm-up solve, the number of iterations, and the nodes and edges of the solver graph. Each graph and setting was timed once. The same measurement reappears as the full-cohort row of Table 4.2 and as the $d = 0.85$ row of Table 4.6, and Table 8.2 in Appendix 8.1 lists every timed run.

Table 4.1: Computational benchmark on the full graphs of the matched cohort ($n = 15{,}107$ contracts; AWP/EndorseRank runtime ratio 2.2× at an edge ratio of 2.2×). Runtime = mean wall-clock seconds over 5 timed solver calls after one untimed warm-up, on which alone peak memory is traced; SD over the same 5 runs. A timed call turns an edge list into scores: node indexing, sparse-matrix assembly, row normalisation, restart vector and power iteration. $|V|$ and $|E|$ are the nodes and distinct edges of each method's solver graph. C-PR walks on the union of the allowance layer and the transfer layer, so its $|E|$ counts the distinct pairs of both; S-PR runs on the transfer layer, and its runtime and iterations include the solve of the allowance layer that supplies its restart vector. Table 8.2 lists every run.

Method	Runtime (s)	SD (s)	Peak mem. (MB)	Iters	$\lvert V\rvert$	$\lvert E\rvert$
EndorseRank	10.294	0.601	879.7	69	6,219,063	11,210,114
AWP	22.367	1.825	1,712.6	92	7,882,356	25,001,216
C-PR ($\lambda = 0.5$)	36.162	2.112	1,768.1	79	9,151,010	31,552,359
S-PR	34.422	3.833	1,235.8	158	7,884,602	25,018,892

EndorseRank solves the allowance graph (6,219,063 nodes, 11,210,114 edges) in 10.29 s on average

and AWP solves the transfer graph (7,882,356 nodes, 25,001,216 edges) in 22.37 s. AWP thus takes 2.17 times as long as EndorseRank, at an edge ratio of 2.23. Assembling the sparse matrix and running each power iteration are both linear in $|E|$ (Appendix 8.3.3), and the time ratio stays close to the edge ratio, slightly below it, although AWP needs more iterations (92 against 69, as in the evaluation runs at T_{obs}). Peak memory is 879.7 MB for EndorseRank and 1,712.6 MB for AWP, and the standard deviations over the five timed runs are 0.60 s and 1.83 s. The speed is a consequence of $|E|$: on the same contracts the allowance graph has 44.8% of the transfer graph's edges, and both scores use the same solver.

The two hybrid rows show what the speed result covers and where it stops. C-PR walks on the union of both edge sets (9,151,010 nodes, 31,552,359 edges) and needs 36.16 s, 1.62 times as long as AWP on 1.26 times its edges. S-PR runs on the transfer layer (7,884,602 nodes, 25,018,892 edges) and takes 34.42 s, 1.54 times as long as AWP. Both its time and its iteration count include the solve of the allowance layer that supplies its restart vector; in the evaluation run at T_{obs} its 158 iterations are 63 for that solve and 95 for the seeded transfer walk. The lower cost is thus confined to the score that walks the allowance graph alone and comes from that graph's size: any algorithm that walks the transfer layer, on its own or combined with allowance information, pays for the edges of that layer.

Two caveats attach to this result. The time covers only the solver call on edge lists built in advance; a full refresh would add parsing, decoding and edge-list construction, which all algorithms share wholly or in part. And a faster method is not necessarily a better one, so the runtime has to be read against the alignment and holdout results that follow.

4.2.1 Runtime on subsamples of the matched cohort

Table 4.2 reports the runtime of EndorseRank and AWP at each stage of Section 4.1.1, together with the ratio of the two runtimes, the ratio of the two edge counts and the nodes and edges of both solver graphs at that stage. The stages change the edge ratio as well as the size. The transfer graph has 3.98 times as many edges as the allowance graph at $n = 1{,}000$ (1,692,513 against 424,888),

2.49 times at $n = 2{,}000$, 2.60 times at $n = 5{,}000$, 2.57 times at $n = 10{,}000$ and 2.23 times on the full cohort. If the time of a solve follows $|E|$, the time ratio should follow these edge ratios; it is 2.89, 4.55, 3.60, 2.70 and 2.17 at the five stages. From $n = 10{,}000$ on, the time ratio stays within 0.13 of the edge ratio, and on the full cohort it lies just below it. At $n = 5{,}000$ it exceeds the edge ratio by 1.00, and at the two smallest stages, where EndorseRank's solve takes less than a second, it is 0.73 and 1.83 times the edge ratio. The time ratio thus follows the edge ratio on the two largest graphs but not on the three smaller ones. Figure 4.1 plots both runtimes against the stage size, with the edge count of each graph beside its point. The stages are subsamples of one cohort graph, so they show how runtime grows with the edges up to the full graph and not beyond it.

Table 4.2: Runtime of EndorseRank (ER) and AWP against the size of the evaluation set: deterministic SHA256 subsamples of the matched cohort (seed `contract-benchmark-v1`) up to the full cohort of $n = 15{,}107$ contracts. Each stage keeps the extracted edges incident to its sampled contracts, so $|V|$ and $|E|$ of both solver graphs are reported per stage. AWP/ER time and AWP/ER $|E|$ are the ratios of the two runtimes and of the two edge counts in the same row. Runtime = mean wall-clock seconds over 5 timed solver calls after one untimed warm-up. [a]The full-cohort row is the measurement of Table 4.1, reused, not timed again.

n	ER (s)	AWP (s)	AWP/ER time	AWP/ER $\|E\|$	ER $\|V\|$	ER $\|E\|$	AWP $\|V\|$	AWP $\|E\|$
1,000	0.580	1.677	2.9×	4.0×	404,563	424,888	1,124,621	1,692,513
2,000	0.939	4.272	4.6×	2.5×	1,112,404	1,263,471	1,917,597	3,148,826
5,000	2.496	8.995	3.6×	2.6×	2,104,606	2,988,450	3,539,528	7,769,268
10,000	6.220	16.821	2.7×	2.6×	4,291,078	6,673,167	6,247,634	17,180,184
15,107[a]	10.294	22.367	2.2×	2.2×	6,219,063	11,210,114	7,882,356	25,001,216

4.3 Same-window alignment (Claim C2)

Claim C2 asks what each score agrees with in the window it is computed from. Table 4.3 reports the family-mean Kendall τ of every score on the transfer, allowance and Sybil-stability families, each with a 95% paired-bootstrap interval. The raw degrees have no row because in this window they are proxies: in-degree belongs to the transfer family and in-approve degree to the allowance family. The secondary part of rule A (Section 3.7) is the only same-window comparison that the plan fixed; here it is applied to the revised scores, after the registered results were known, so its

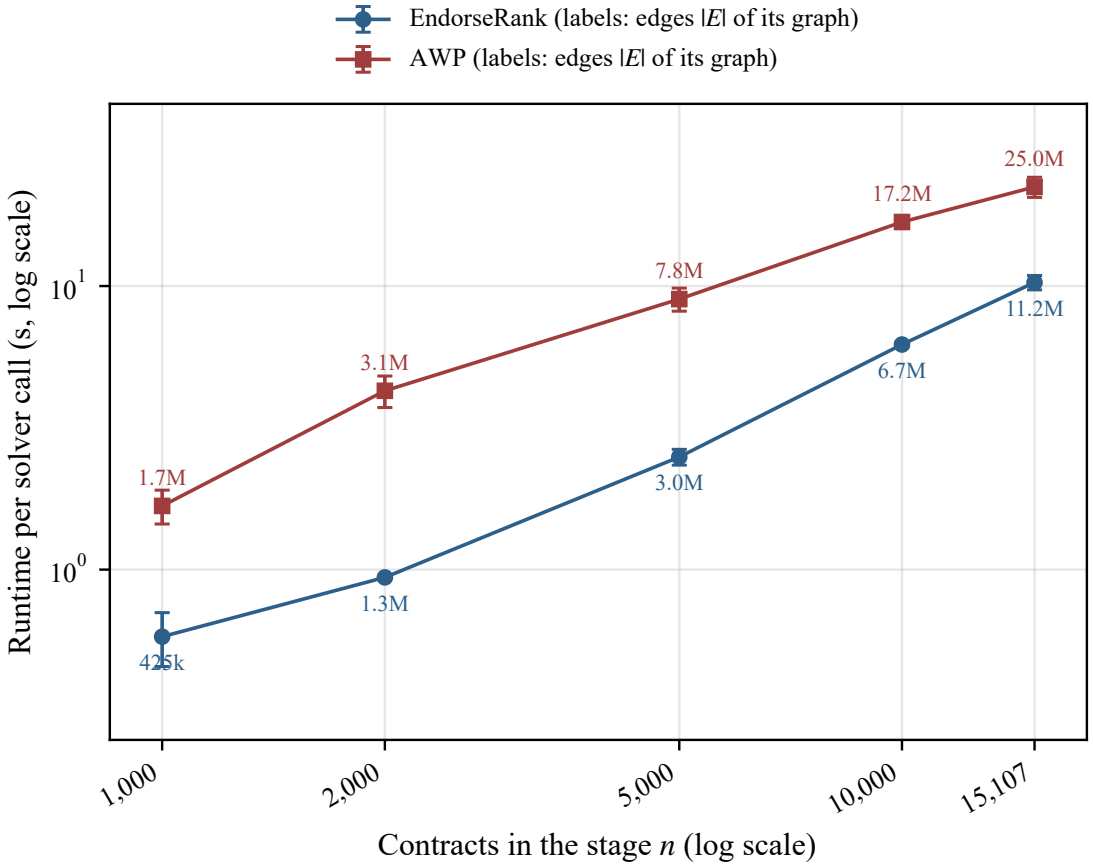


Figure 4.1: Wall-clock time of one solver call of EndorseRank and AWP against the number of contracts in the stage, both axes on a log scale. The stages are the deterministic SHA256 subsamples of the matched cohort (Table 4.2); each keeps the extracted edges incident to its contracts, and the label beside each point gives the edge count $|E|$ of that graph. The rightmost point is the full cohort of 15,107 contracts. Error bars are the standard deviations over five timed runs.

verdict is a re-analysis. The paired contrasts below were not named in the plan and were computed together with the same-window results.

Table 4.3: Same-window family-mean Kendall τ with bootstrap intervals for every score of this study (matched cohort, $n = 15{,}107$ contracts). The scores are EndorseRank and AWP, the coupled operator C-PR at five values of λ and the endorsement-seeded walk S-PR; $\lambda = 1$ is the allowance layer walked alone (latest allowances in USD, each capped) and $\lambda = 0$ the transfer layer walked alone (USD amounts discounted by the decay of AWP); both restart uniformly. The raw degrees have no row because they are proxies of the transfer and allowance families; EndorseRank with activity restarts is in Table 4.9. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Method	Transfer	Allowance	Sybil
EndorseRank	0.158 [0.147, 0.169]	0.585 [0.578, 0.592]	0.007 [-0.001, 0.015]
AWP	0.572 [0.564, 0.580]	0.193 [0.182, 0.203]	0.395 [0.387, 0.403]
C-PR ($\lambda = 1$)	0.162 [0.151, 0.173]	0.597 [0.590, 0.604]	0.011 [0.002, 0.019]
C-PR ($\lambda = 0$)	0.659 [0.653, 0.666]	0.225 [0.214, 0.235]	0.404 [0.396, 0.413]
C-PR ($\lambda = 0.25$)	0.385 [0.374, 0.395]	0.457 [0.449, 0.465]	0.155 [0.147, 0.164]
C-PR ($\lambda = 0.5$)	0.365 [0.355, 0.375]	0.477 [0.469, 0.485]	0.142 [0.134, 0.151]
C-PR ($\lambda = 0.75$)	0.344 [0.333, 0.354]	0.492 [0.484, 0.500]	0.129 [0.121, 0.138]
S-PR	0.629 [0.621, 0.636]	0.323 [0.313, 0.333]	0.398 [0.390, 0.406]

The scores divide by the edges they walk on. EndorseRank reaches 0.585 $[0.578, 0.592]$ on the allowance family and 0.158 $[0.147, 0.169]$ on the transfer family; AWP reaches 0.572 $[0.564, 0.580]$ on the transfer family and 0.193 $[0.182, 0.203]$ on the allowance family. Each score is ahead of the other on the family of its own edges, EndorseRank by $+0.392$ $[0.380, 0.405]$ on the allowance family and AWP by $+0.414$ $[0.401, 0.427]$ on the transfer family (Table 8.7 in Appendix 8.5). The two layers of C-PR walked alone repeat the split: C-PR ($\lambda = 1$) reaches 0.597 $[0.590, 0.604]$ on the allowance family and C-PR ($\lambda = 0$) 0.659 $[0.653, 0.666]$ on the transfer family. On the Sybil-stability family the scores that walk on transfers lead, AWP with 0.395 $[0.387, 0.403]$ against 0.007 $[-0.001, 0.015]$ for EndorseRank. Part of this agreement is built in. The main proxies of each family come from the same events as one of the graphs, in-degree and US-dollar in-value from the transfers of the five tokens and in-approve degree and capped value from their latest allowances, and a PageRank smooths the in-degree of its own graph. Agreement with the proxies of a score's

own edge type is therefore an intended-construct check and cannot count as an external test. The holdouts of Section 4.6 ask the same question out of window.

Appendix 8.5.2 breaks the family means of EndorseRank and AWP into their proxies. AWP follows the summed US-dollar value of the inbound transfers (0.566 $[0.558, 0.575]$) almost as closely as their number of senders, the in-degree (0.578 $[0.569, 0.587]$). That is what its weights lead one to expect. With the slope set at the median transfer (Section 3.6.1), a transfer of 200 dollars weighs one half, one of 400 dollars 0.8 and one of 100 dollars 0.27, so an edge weight grows with the amounts transferred as well as with their number. In the registered analysis, with $b = 1$ on raw base units, V was close to one for nearly every transfer whatever its amount, and AWP followed the in-degree (0.541 $[0.534, 0.547]$) much more closely than the in-value (0.353 $[0.345, 0.361]$; registered results digest, Appendix 8.6.10). On the Sybil-stability family AWP follows transfer tenure (0.508 $[0.499, 0.517]$) and active months (0.549 $[0.540, 0.558]$) and is positive on the inbound-counterparty ratio (0.127 $[0.114, 0.141]$).

EndorseRank follows the in-approve degree (0.655 $[0.648, 0.662]$) more closely than the in-approve value (0.515 $[0.506, 0.524]$). Its value weight is 0.8 at twice the median allowance of 62 dollars and close to one at a few hundred dollars, so such an allowance and an unlimited one weigh nearly alike, whereas the in-approve value counts each allowance up to the cap of 20,000 dollars. Its transfer coefficients are 0.182 $[0.169, 0.194]$ with in-degree and 0.134 $[0.123, 0.145]$ with in-value. On the Sybil-stability family it is negative on the inbound-counterparty ratio (-0.156 $[-0.166, -0.147]$) and reaches 0.080 $[0.069, 0.091]$ on tenure and 0.098 $[0.087, 0.109]$ on active months, so how many owners approve a contract says little about how long it has received transfers. C-PR ($\lambda = 1$) walks the same allowance graph with capped US-dollar amounts and also restarts uniformly; it stays close to EndorseRank on every family, and its point estimate on the in-approve value, which sums the same capped amounts as its own layer does, is higher (0.540 $[0.531, 0.549]$ against 0.515 $[0.506, 0.524]$; no paired contrast was computed).

Because every contract in the cohort received approvals of the five tokens from at least three owners, EndorseRank leaves few ties here. It gives 13,625 distinct scores to the 15,107 contracts, and only

0.17% of contract pairs are tied, most of them among the 535 contracts that score zero (Section 4.4). Its coefficients therefore rest on an ordering of nearly every contract, and keeping the contracts outside a graph as isolated nodes changes none of them (Table 4.5).

Claim C2 is supported as an intended-construct check, with agreement that is partly built in. Every score agrees most with the proxies of the edge type it walks on, with intervals far from zero, and the mixtures of C-PR lie between their two layers on every family (Section 4.3.1).

4.3.1 The coupled operator (RQ5)

Two hybrids were defined in Section 3.6.3. The coupled operator C-PR mixes the US-dollar allowance and transfer layers in each node's transition row with weight λ, and the endorsement-seeded walk S-PR keeps the transfer layer but restarts from the allowance walk. Table 4.4 gives their paired contrasts in this window: first eight against the two layers walked alone, then four against EndorseRank and AWP. RQ5 is decided out of window (Sections 4.6 and 4.6.2). Here we only locate the hybrids on the checks that the single-layer scores were built to pass.

C-PR lies between its two layers on every family. At $\lambda = 0.5$ it reaches 0.365 $[0.355, 0.375]$ on the transfer family, 0.477 $[0.469, 0.485]$ on the allowance family and 0.142 $[0.134, 0.151]$ on the Sybil-stability family, and as λ rises from 0.25 to 0.75 it moves toward the allowance layer on all three (Table 4.3). It stays nearer to its allowance layer than to its transfer layer: on the allowance family it falls 0.120 below C-PR ($\lambda = 1$) ($[-0.127, -0.113]$), and on the transfer family 0.294 below C-PR ($\lambda = 0$) ($[-0.305, -0.284]$). The mixing rule of Equation (3.9) acts at every node with out-edges in both layers. Almost every node of the allowance layer is an owner with out-edges, so the allowance rows enter the walk wherever an owner also sent a transfer that the extraction kept. Against the single-layer scores C-PR is below AWP on the transfer family (-0.207, $[-0.220, -0.194]$) and below EndorseRank on the allowance family (-0.108, $[-0.115, -0.101]$). On the Sybil-stability family it lies above the allowance walk ($+0.131$, $[0.126, 0.137]$) and EndorseRank ($+0.135$, $[0.129, 0.141]$) and below the transfer walk (-0.262, $[-0.273, -0.253]$) and AWP (-0.253, $[-0.263, -0.242]$).

On every family S-PR falls below the single layer that leads that family. Restarting the transfer

Table 4.4: Same-window paired contrasts for the coupled operator (C-PR, $\lambda = 0.5$) and the endorsement-seeded walk (S-PR) against the two layers they are built from, the allowance layer (C-PR ($\lambda = 1$)) and the transfer layer (C-PR ($\lambda = 0$)), and of C-PR against EndorseRank and AWP (matched cohort, $n = 15{,}107$ contracts). Family-mean τ; the same contract resamples are used for a and b. The contrasts with the layers are a stricter check beside the secondary part of rule A, which compares C-PR's point estimate with the better layer's interval in Table 4.3. The secondary part of rule A, fixed in the analysis plan and applied here to the revised scores, is not met: C-PR's point estimate lies outside the interval of C-PR ($\lambda = 0$) on the transfer family and outside the interval of C-PR ($\lambda = 1$) on the allowance family, and C-PR does not exceed both layers on the Sybil-stability family. An interval below 0 means a is below b beyond sampling variation; one above 0 means it exceeds it. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Contrast (a minus b)	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
Against the two layers walked alone					
C-PR transfer minus C-PR ($\lambda = 0$) transfer	0.365	0.659	-0.294	[-0.305, -0.284]	0.000
C-PR allowance minus C-PR ($\lambda = 1$) allowance	0.477	0.597	-0.120	[-0.127, -0.113]	0.000
C-PR sybil-stability minus C-PR ($\lambda = 0$) sybil-stability	0.142	0.404	-0.262	[-0.273, -0.253]	0.000
C-PR sybil-stability minus C-PR ($\lambda = 1$) sybil-stability	0.142	0.011	+0.131	[0.126, 0.137]	1.000
S-PR transfer minus C-PR ($\lambda = 0$) transfer	0.629	0.659	-0.031	[-0.035, -0.028]	0.000
S-PR allowance minus C-PR ($\lambda = 1$) allowance	0.323	0.597	-0.274	[-0.285, -0.263]	0.000
S-PR sybil-stability minus C-PR ($\lambda = 0$) sybil-stability	0.398	0.404	-0.007	[-0.009, -0.004]	0.000
S-PR sybil-stability minus C-PR ($\lambda = 1$) sybil-stability	0.398	0.011	+0.387	[0.376, 0.399]	1.000
Against EndorseRank and AWP					
C-PR transfer minus AWP transfer	0.365	0.572	-0.207	[-0.220, -0.194]	0.000
C-PR allowance minus EndorseRank allowance	0.477	0.585	-0.108	[-0.115, -0.101]	0.000
C-PR sybil-stability minus AWP sybil-stability	0.142	0.395	-0.253	[-0.263, -0.242]	0.000
C-PR sybil-stability minus EndorseRank sybil-stability	0.142	0.007	+0.135	[0.129, 0.141]	1.000

walk from the allowance walk lowers the transfer coefficient to 0.629 $[0.621, 0.636]$ ($\Delta\tau = -0.031$, $[-0.035, -0.028]$, against C-PR ($\lambda = 0$)) and the Sybil-stability coefficient to 0.398 $[0.390, 0.406]$ (-0.007, $[-0.009, -0.004]$). It raises the allowance coefficient to 0.323 $[0.313, 0.333]$, against 0.225 $[0.214, 0.235]$ for the transfer walk with uniform restarts (no paired contrast was computed for this pair), yet leaves it 0.274 below the allowance walk ($[-0.285, -0.263]$). A restart vector changes where the walk lands after each teleport. It carries part of the allowance signal into the transfer walk, but less than walking the allowance edges does.

As fixed in the analysis plan, the secondary part of rule A asks C-PR's point estimate at $\lambda = 0.5$ to lie within the 95% interval of the better layer on the transfer and allowance families, and C-PR to exceed both layers on the Sybil-stability family. Applied to the revised scores, all three parts fail, as they did in the registered analysis (Table 8.13). The transfer coefficient (0.365) lies below the interval of C-PR ($\lambda = 0$) ($[0.653, 0.666]$), the allowance coefficient (0.477) lies below that of C-PR ($\lambda = 1$) ($[0.590, 0.604]$), and on the Sybil-stability family C-PR exceeds only the allowance layer. The paired contrasts of Table 4.4, a stricter check that is not part of the rule, point the same way. In the same window the mixture gives up part of each layer's agreement with its own family and exceeds the better layer on no family. The holdout takes up the remaining question of whether it has predictive content that the single layers lack.

4.4 Rank divergence (H1 / RQ1)

Hypothesis H1 and RQ1 concern whether EndorseRank and AWP put the same contracts in different orders. The inter-method Kendall τ_b between the two score vectors is 0.174, with interval $[0.163, 0.185]$ (last row of Table 4.5), so the interval excludes both zero and one. The two rankings are related, yet far from interchangeable. Of the 114,103,171 contract pairs, 96.7% are ordered by both scores, and 41.2% of these are ordered in opposite directions.

Ties play almost no part in this result. The marginal score distributions are in Figure 4.2. EndorseRank gives 13,625 distinct scores to the 15,107 contracts and ties 0.17% of the pairs; AWP

gives 12,435 distinct scores and ties 3.13%. Under each score the largest tie block is formed by the contracts that score zero: 535 under EndorseRank and 2,672 under AWP. They are the 523 and 2,670 contracts outside each graph, 80 of them outside both, and 12 and 2 contracts whose only pairs carry amounts too small for the value weight to differ from zero. Under EndorseRank 166 further contracts share one score, $c = 7.64 \times 10^{-8}$, the score of a node without in-edges; they are in the allowance graph only as owners. Every other cohort contract has received at least one positive allowance and scores above c. A marginal distribution cannot show how two scores order the same contracts, so Figure 4.3 plots the two rankings against each other for a random sample of 3,000 contracts, with tied contracts at their mid-rank as in τ_b. The zero blocks appear as a column at the right edge (EndorseRank) and a row at the top (AWP). Contracts that both scores rank near the top, or both near the bottom, lie close to the identity line; in between, the points spread over most of the square.

The zero blocks come from one rule of the pipeline: a contract missing from a graph scores zero (Section 3.6). The analysis plan fixed the alternative, keeping missing contracts as isolated nodes, as a sensitivity analysis, and Table 4.5 gives its result on the revised scores. No coefficient changes at the third decimal, and the agreement between the two scores stays at 0.174 $[0.163, 0.185]$. The reason is where the missing contracts land. Under EndorseRank's uniform restarts an isolated node receives c, the lowest score that any node of the graph can have, so the 523 contracts outside the allowance graph stay near the bottom of the order and join the 166 contracts at c. The 12 contracts whose only pairs carry zero weight are nodes of the allowance graph, so the rule does not add them, and the solver, which keeps only edges of positive weight, leaves them out; they keep the score zero. No pair changes direction: the pairs between the 523 and the 166 become ties, and the pairs between the 523 and the 12, tied at zero under the main rule, become ordered. Under AWP's restarts an isolated contract that sent nothing receives no restart mass and scores zero under both rules. Because the cohort was defined by approvals received, nearly every cohort contract is a spender in the allowance graph, and the rule for missing contracts decides almost nothing.

H1 is supported. The two scores order the cohort differently, and the difference does not come from

Table 4.5: Same-window family-mean Kendall τ with contracts missing from a graph scored zero or kept as isolated nodes (matched cohort, $n = 15{,}107$ contracts). The main analysis scores a contract that is missing from a method's graph zero; keeping it as an isolated node is the sensitivity analysis fixed in the analysis plan. The rule concerns 523 contracts for EndorseRank and 2,670 for AWP (80 are missing from both graphs). Under EndorseRank's uniform restarts an isolated node gets the score of a node without in-edges. AWP restarts only at addresses that sent a transfer, so an isolated contract keeps the score zero and AWP's coefficients are the same under both rules. The last row is the agreement between the two scores. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Missing contracts scored zero		Missing contracts as isolated nodes	
Family	EndorseRank	AWP	EndorseRank	AWP
Transfer	0.158 [0.147, 0.169]	0.572 [0.564, 0.580]	0.158 [0.146, 0.169]	0.572 [0.564, 0.580]
Allowance	0.585 [0.578, 0.592]	0.193 [0.182, 0.203]	0.585 [0.579, 0.592]	0.193 [0.182, 0.203]
Sybil stability	0.007 [-0.001, 0.015]	0.395 [0.387, 0.403]	0.007 [-0.002, 0.015]	0.395 [0.387, 0.403]
EndorseRank vs. AWP	0.174 [0.163, 0.185]		0.174 [0.163, 0.185]	

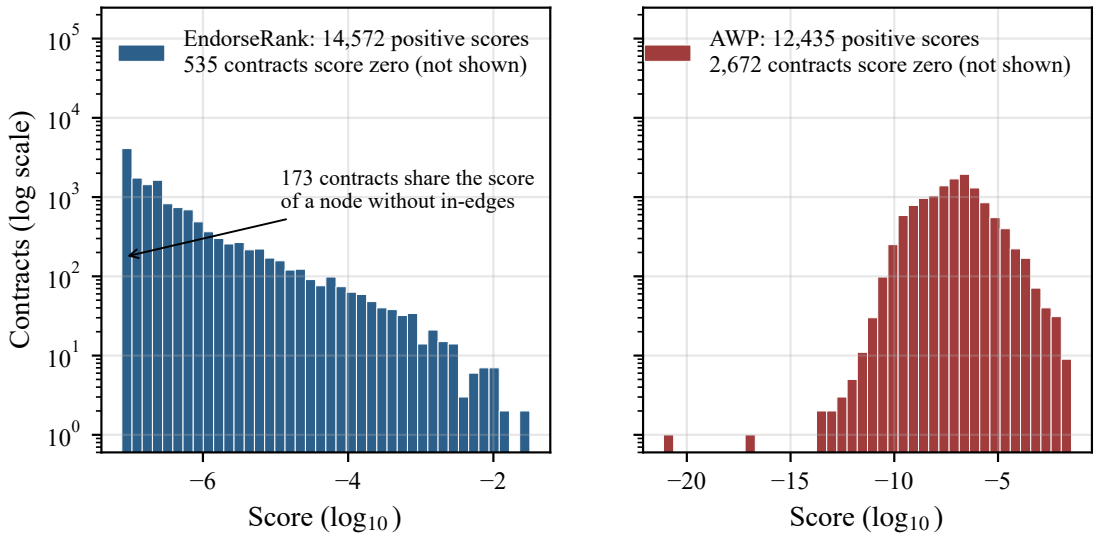


Figure 4.2: Marginal EndorseRank and AWP score distributions on the matched cohort ($n = 15{,}107$ contracts), drawn on a $\log_{10}$ scale for contracts with a positive score; counts are on a log scale. Zero-score contracts, which no edge of positive weight in the respective graph touches, are left out: 535 for EndorseRank and 2,672 for AWP. For EndorseRank, the 166 contracts that are in the allowance graph only as owners share the score $c = 7.64 \times 10^{-8}$ of a node without in-edges, the lowest positive score.

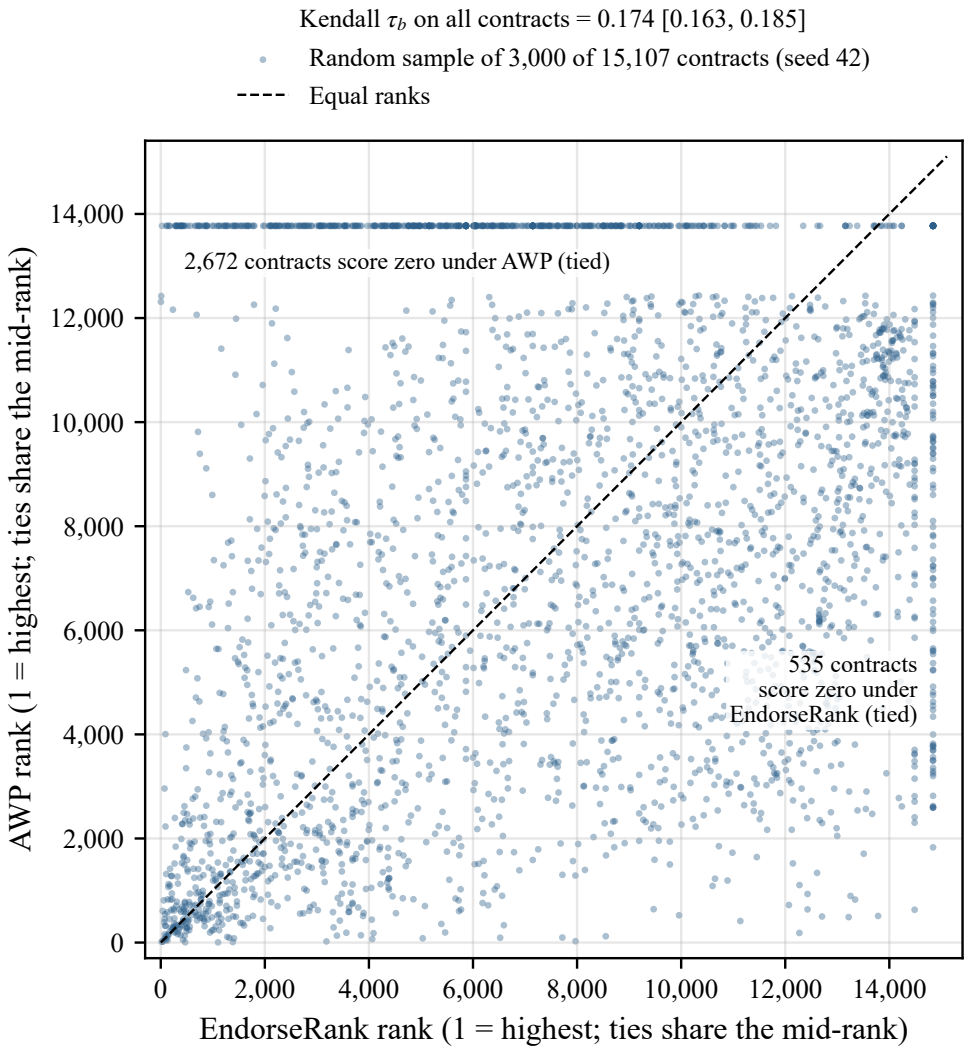


Figure 4.3: Rank–rank scatter of EndorseRank against AWP on the matched cohort (random sample of 3,000 of the 15,107 contracts, seed 42). Tied scores share their average rank, which is the tie treatment of Kendall's τ_b. The column at the right edge holds the sampled contracts among the 535 that score zero under EndorseRank and the row at the top those among the 2,672 that score zero under AWP. A point on the dashed identity line has the same rank in both orderings; the legend gives τ_b on all contracts with its 95% interval.

ties or from the rule for missing contracts. A protocol that relies on transfer-based scores cannot assume that its ordering of contracts survives a switch to allowance-based scores or a blend with them.

4.5 Robustness and sensitivity checks

Three checks, run for EndorseRank and AWP, vary the damping factor, the slope of the value weight and the sample definition. The analysis plan fixed the damping values; the value-slope check takes the place of the plan's token check, which lost its point once every graph keeps five tokens (Section 3.12.3), and the sample-definition check was not in the plan. All three were run on the revised scores after the registered results were known. The first two checks hold the sample fixed and ask whether the results hinge on one setting. The third changes the sample itself, so its coefficients are expected to move, and it is not evidence of robustness. A fourth check asks how much EndorseRank depends on its restart rule.

4.5.1 Damping

Table 4.6 recomputes the family means at $d \in \{0.75, 0.85, 0.95\}$ on the same graphs.

Table 4.6: Robustness: family-mean Kendall τ of EndorseRank (ER) and AWP under PageRank damping $d \in \{0.75, 0.85, 0.95\}$ on the matched cohort ($n = 15{,}107$ contracts); all. = allowance, tr. = transfer, Syb. = Sybil stability. Runtimes are means of five timed solver calls on the full graphs; the $d = 0.85$ runtimes are the measurements of Table 4.1, reused, not timed again (Table 8.2 lists every run). CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

d	ER all.	AWP all.	ER tr.	AWP tr.	ER Syb.	AWP Syb.	ER (s)	AWP (s)
0.75	0.586 [0.580, 0.594]	0.193 [0.182, 0.203]	0.158 [0.146, 0.168]	0.570 [0.562, 0.578]	0.007 [-0.002, 0.015]	0.397 [0.389, 0.406]	5.984	15.236
0.85	0.585 [0.578, 0.592]	0.193 [0.182, 0.203]	0.158 [0.147, 0.169]	0.572 [0.564, 0.580]	0.007 [-0.001, 0.015]	0.395 [0.387, 0.403]	10.294	22.367
0.95	0.583 [0.577, 0.591]	0.192 [0.181, 0.202]	0.158 [0.147, 0.169]	0.577 [0.569, 0.585]	0.008 [-0.001, 0.016]	0.391 [0.384, 0.400]	31.233	60.962

For EndorseRank no family mean moves by more than 0.003 across the three values; its allowance coefficient is 0.586 $[0.580, 0.594]$ at $d = 0.75$ and 0.583 $[0.577, 0.591]$ at $d = 0.95$. AWP moves

most on the transfer family, from 0.570 $[0.562, 0.578]$ to 0.577 $[0.569, 0.585]$, and its Sybil-stability coefficient falls from 0.397 $[0.389, 0.406]$ to 0.391 $[0.384, 0.400]$. Neither score's pattern is an artifact of the conventional $d = 0.85$. EndorseRank's stability is largely structural. For a spender whose owners receive no approvals, EndorseRank equals $c\,(1 + d\,\delta(v))$, with $\delta(v)$ the owner-normalized in-strength (Section 5.3.2), and the order that this formula gives among such spenders does not depend on d; 12,039 of the 14,406 cohort contracts with in-edges are spenders of this kind. The number of iterations grows with d, from 41 to 69 and 217 for EndorseRank and from 55 to 92 and 261 for AWP. The runtimes are 5.98, 10.29 and 31.23 s for EndorseRank and 15.24, 22.37 and 60.96 s for AWP, time ratios of 2.55, 2.17 and 1.95 at an edge ratio of 2.23, on the full graphs of Table 4.1 (6,219,063 nodes and 11,210,114 edges for EndorseRank, 7,882,356 and 25,001,216 for AWP); each graph and damping pair was timed once, and the $d = 0.85$ values are the main measurement, reused. Figure 4.4 plots the family means with their intervals.

4.5.2 Value-slope sensitivity

Table 4.7 recomputes EndorseRank and AWP at T_{obs} with the slope b of the value weight divided and multiplied by ten (Section 3.12.3), on the same graphs and proxies.

Table 4.7: Robustness: family-mean Kendall τ on the matched cohort ($n = 15{,}107$ contracts) when the slope b of the value weight V is divided or multiplied by ten, next to the main analysis. The main slopes put $V = 1/2$ at the median amount, \$200 for transfers and \$62 for allowances; $b/10$ moves that point to ten times the median and $10b$ to a tenth of it. The rank agreement τ of each variant with the main score of the same method (EndorseRank and AWP) is 0.973 and 0.973 at $b/10$, and 0.968 and 0.971 at $10b$. No intervals were computed for the variants.

Method	Transfer			Allowance			Sybil		
	$b/10$	b	$10b$	$b/10$	b	$10b$	$b/10$	b	$10b$
EndorseRank	0.157	0.158	0.159	0.588	0.585	0.580	0.006	0.007	0.009
AWP	0.577	0.572	0.567	0.193	0.193	0.191	0.397	0.395	0.393

No family mean moves by more than 0.0054, and the pattern stays: EndorseRank leads on the allowance family, and AWP on the transfer and Sybil-stability families. The steeper slope $10b$ puts the half weight at a tenth of the median amount, so that most amounts weigh close to one and an

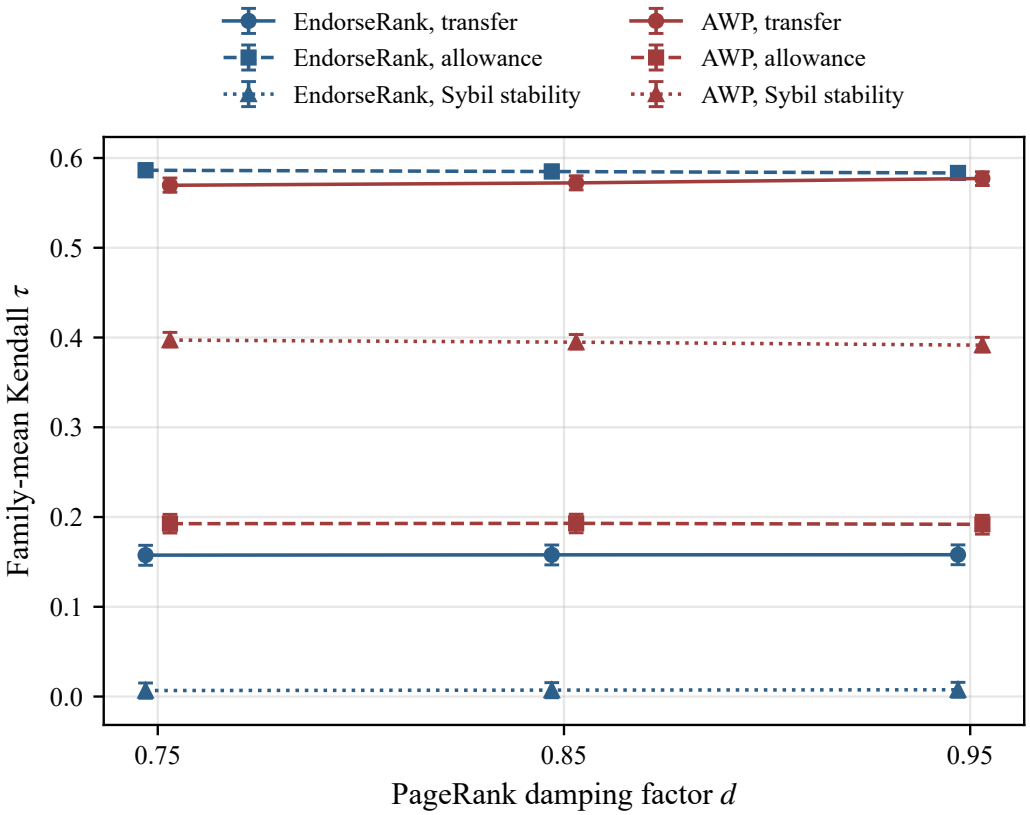


Figure 4.4: Damping sensitivity of the family-mean Kendall τ of EndorseRank and AWP on the transfer, allowance and Sybil-stability families across $d \in \{0.75, 0.85, 0.95\}$, with 95% bootstrap intervals (values from Table 4.6; matched cohort, $n = 15{,}107$ contracts).

edge weight comes nearer to a count of events. It lowers EndorseRank's allowance coefficient from 0.585 to 0.580 and AWP's transfer coefficient from 0.572 to 0.567; the flatter slope $b/10$ raises them to 0.588 and 0.577. Each variant orders the contracts almost as the main score of the same method does, at $\tau = 0.973$ for both scores at $b/10$, and at 0.968 for EndorseRank and 0.971 for AWP at $10b$. No intervals were computed for the variants. The family-level results therefore do not hinge on the amount at which the value weight reaches one half.

4.5.3 Sample-definition sensitivity

Table 4.8 recomputes the family means on the subsamples of Section 4.1.1, each scored on the edges incident to its contracts.

Table 4.8: Sensitivity of the family means to the evaluation set: family-mean Kendall τ of EndorseRank (ER) and AWP on deterministic SHA256 subsamples of the matched cohort (seed `contract-benchmark-v1`, full cohort $n = 15{,}107$ contracts), each stage scored on the edges incident to its contracts; all. = allowance, tr. = transfer, Syb. = Sybil stability. Rows are evaluated on different contract sets, so their τ values are not comparable with Table 4.3; they show how much the statistic moves with the evaluation population. No intervals were computed for the stages. Runtimes are the measurements of Table 4.2.

n	ER all.	AWP all.	ER tr.	AWP tr.	ER Syb.	AWP Syb.	ER (s)	AWP (s)	AWP/ER time
1,000	0.715	0.196	0.205	0.586	0.038	0.408	0.580	1.677	2.9×
2,000	0.686	0.199	0.196	0.588	0.033	0.411	0.939	4.272	4.6×
5,000	0.634	0.211	0.195	0.571	0.034	0.391	2.496	8.995	3.6×
10,000	0.607	0.196	0.167	0.571	0.013	0.394	6.220	16.821	2.7×
15,107	0.585	0.193	0.158	0.572	0.007	0.395	10.294	22.367	2.2×

The coefficients move with the sample. EndorseRank moves most: its allowance coefficient is 0.715 at $n = 1{,}000$ and falls with every stage to 0.585 on the full cohort, and its transfer coefficient falls from 0.205 to 0.158. AWP stays within 0.018 of its full-cohort values on every family. No intervals were computed for the stages. None of these shifts is evidence about the scores. A stage keeps only the edges incident to its sampled contracts, so the paths that pass through unsampled cohort contracts are cut, and each score comes closer to a count of the sampled contract's own edges, which is what the proxies count. We therefore report the table as a sensitivity analysis of the

sample definition, and its values must not be compared with those of Table 4.3; the matched cohort is the primary sample.

4.5.4 Restarts of EndorseRank

EndorseRank restarts the walk uniformly, while AWP restarts it at addresses in proportion to their recent sending activity (Section 3.6.2). Carried over to allowances, AWP's rule restarts the walk at owners in proportion to the spenders they approved, each weighted by the age of the latest approval. The analysis plan fixed uniform restarts for EndorseRank and named the variant with AWP's restarts as a score to report beside it, without fixing any comparison between the two. Table 4.9 scores both versions, with AWP beside them.

Table 4.9: EndorseRank with uniform restarts and with restarts weighted by approving activity, AWP's rule applied to the allowance graph, next to AWP: family-mean Kendall τ on the matched cohort ($n = 15{,}107$ contracts) and τ with the two spender labels in W0 ($n = 17{,}050$ spender contracts, scores frozen at 31 March 2026) and W1 ($n = 18{,}152$, frozen at 30 June 2026). Both EndorseRank rows weight each latest allowance by $\sigma(\Delta t)V(z)$. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

		Same window, matched cohort			W0, spenders		W1, spenders	
Method	Restarts	Transfer	Allowance	Sybil	New approval pairs	New senders	New approval pairs	New senders
EndorseRank	uniform	0.158 [0.147, 0.169]	0.585 [0.578, 0.592]	0.007 [-0.001, 0.015]	0.231 [0.220, 0.242]	0.200 [0.189, 0.210]	0.234 [0.222, 0.244]	0.194 [0.184, 0.205]
EndorseRank	by approving activity	0.164 [0.153, 0.173]	0.426 [0.418, 0.435]	0.049 [0.039, 0.057]	0.308 [0.299, 0.319]	0.269 [0.260, 0.278]	0.295 [0.285, 0.304]	0.242 [0.232, 0.253]
AWP	by sending activity	0.572 [0.564, 0.580]	0.193 [0.182, 0.203]	0.395 [0.387, 0.403]	0.196 [0.181, 0.210]	0.346 [0.336, 0.356]	0.169 [0.157, 0.183]	0.330 [0.321, 0.340]

With activity restarts, EndorseRank's same-window allowance coefficient falls from 0.585 $[0.578, 0.592]$ to 0.426 $[0.418, 0.435]$, and its transfer coefficient changes little (0.164 $[0.153, 0.173]$ against 0.158 $[0.147, 0.169]$). The restart mass then goes to the owners that approve most, so the score weights a spender by how active its owners are as well as by how many there are. Out of window the variant ranks both spender labels higher in both windows. In W0 it reaches 0.308 $[0.299, 0.319]$ on new approval pairs and 0.269 $[0.260, 0.278]$ on new senders, $+0.077$ $[0.071, 0.084]$ and $+0.070$ $[0.064, 0.076]$ above the uniform version. In W1 it reaches 0.295 $[0.285, 0.304]$ and 0.242 $[0.232, 0.253]$, $+0.061$ $[0.054, 0.067]$ and $+0.048$ $[0.042, 0.053]$ above it. These contrasts

were computed after the labels of each window were known and lie outside every rule. The uniform rule keeps the allowance construct in the same window, and the activity rule ranks later arrivals better: in W0 the variant ranks new approval pairs above the in-approve degree ($+0.059$, $[0.050, 0.067]$), where the uniform version falls below it (Section 4.6).

4.6 Temporal holdout (RQ4, RQ5)

A same-window check cannot separate a score from its own construct. The holdout instead tests whether centrality measured up to a cut-off predicts behavior after it. Following Section 3.7, scores on the W0 spender cohort are frozen at t_1, and two labels are then counted for each contract between 1 April and 30 June 2026: new owner–spender approval pairs (the endorsement construct) and new transfer senders (the flow construct). Table 4.10 sets every score computed at t_1, the two raw-degree baselines included, against both labels. Table 4.11 gives the paired contrasts: in its first block the two of rule A, fixed in the analysis plan before any score of the study existed and applied here to the revised scores, and in its second block those computed after the labels were known.

The first finding answers the second part of RQ4, and the answer differs between the two walks. On the transfer side, AWP ranks new transfer senders at $\tau = 0.346$ $[0.336, 0.356]$, above the transfer in-degree at t_1 (0.303 $[0.292, 0.315]$), with a paired difference of $+0.043$ $[0.038, 0.048]$. Propagation with AWP's weights and activity restarts thus added information to the count on this label. The transfer layer walked with US-dollar amounts and uniform restarts does not clearly do so: C-PR ($\lambda = 0$) reaches 0.296 $[0.285, 0.307]$, slightly below the in-degree (no paired contrast was computed). On the allowance side the count wins. The in-approve degree at t_1 ranks new approval pairs at 0.250 $[0.237, 0.262]$, EndorseRank at 0.231 $[0.220, 0.242]$ (paired difference -0.019, $[-0.026, -0.011]$) and C-PR ($\lambda = 1$) at 0.223 $[0.213, 0.233]$ (-0.027, $[-0.034, -0.019]$). A spender gains more new owners the more it already has, as preferential attachment would predict (Barabási & Albert, 1999), and the count captures this directly. EndorseRank divides each owner's endorsement among all the spenders that owner approves (Section 5.3.2), so a spender whose owners approve many other

Table 4.10: Temporal holdout W0 on the spender cohort ($n = 17{,}050$ contracts): Kendall τ between scores frozen at 31 March 2026 and two labels counted on the contract between 1 April 2026 and 30 June 2026: the number of new owner–spender approval pairs and the number of addresses that sent a transfer to the contract for the first time. The cohort keeps the 17,050 contracts among the 17,167 spenders with a positive latest allowance at the freeze; 104 EOAs without code and 13 EOAs with an EIP-7702 delegation are left out. 2,070 contracts received at least one new pair and 1,695 at least one new sender. Baseline rows are the raw degrees at t_1 that each single-layer PageRank smooths. C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$) are the allowance and transfer layers of the coupled operator walked alone. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Score at t_1	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
In-approve degree at t_1 (baseline)	0.250 [0.237, 0.262]	0.180 [0.169, 0.192]
Transfer in-degree at t_1 (baseline)	0.135 [0.121, 0.150]	0.303 [0.292, 0.315]
EndorseRank	0.231 [0.220, 0.242]	0.200 [0.189, 0.210]
AWP	0.196 [0.181, 0.210]	0.346 [0.336, 0.356]
C-PR ($\lambda = 1$)	0.223 [0.213, 0.233]	0.188 [0.177, 0.199]
C-PR ($\lambda = 0$)	0.160 [0.144, 0.174]	0.296 [0.285, 0.307]
C-PR ($\lambda = 0.25$)	0.253 [0.243, 0.264]	0.269 [0.257, 0.280]
C-PR ($\lambda = 0.5$)	0.252 [0.241, 0.262]	0.262 [0.250, 0.273]
C-PR ($\lambda = 0.75$)	0.249 [0.238, 0.260]	0.253 [0.242, 0.265]
S-PR	0.172 [0.157, 0.185]	0.295 [0.284, 0.305]

Table 4.11: Paired bootstrap differences on the spender holdout W0 ($n = 17{,}050$ contracts; 400 resamples shared by a and b, seed 42). The first block holds the two contrasts of the primary part of rule A, fixed in the analysis plan committed on 9 October 2026 before the study extracted any data and applied here to the revised scores (C-PR against C-PR ($\lambda = 1$) on new approval pairs and against C-PR ($\lambda = 0$) on new transfer senders). The second block was computed after the labels were known: the increment of C-PR ($\lambda = 1$), the allowance layer walked alone, over its raw degree at t_1, the same comparisons for S-PR, the increments of EndorseRank and AWP over their raw degrees, C-PR against C-PR ($\lambda = 0$) on new approval pairs, and C-PR against EndorseRank and AWP. Primary part of rule A (C-PR no worse than C-PR ($\lambda = 1$) on new approval pairs and better than C-PR ($\lambda = 0$) on new transfer senders): no worse holds, better fails, so the rule is not met. The differences between EndorseRank and AWP are in Table 8.11.

Contrast (a minus b)	Label	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
Rule A, primary part, fixed in the plan						
C-PR minus C-PR ($\lambda = 1$)	new approval pairs	0.252	0.223	+0.029	[0.022, 0.036]	1.000
C-PR minus C-PR ($\lambda = 0$)	new transfer senders	0.262	0.296	-0.034	[-0.040, -0.029]	0.000
Computed after the labels were known						
C-PR ($\lambda = 1$) minus t_1 in-approve degree	new approval pairs	0.223	0.250	-0.027	[-0.034, -0.019]	0.000
S-PR minus C-PR ($\lambda = 1$)	new approval pairs	0.172	0.223	-0.051	[-0.066, -0.038]	0.000
S-PR minus C-PR ($\lambda = 0$)	new transfer senders	0.295	0.296	-0.001	[-0.004, 0.002]	0.265
EndorseRank minus t_1 in-approve degree	new approval pairs	0.231	0.250	-0.019	[-0.026, -0.011]	0.000
AWP minus t_1 in-degree	new transfer senders	0.346	0.303	+0.043	[0.038, 0.048]	1.000
C-PR minus C-PR ($\lambda = 0$)	new approval pairs	0.252	0.160	+0.092	[0.080, 0.105]	1.000
C-PR minus EndorseRank	new approval pairs	0.252	0.231	+0.021	[0.013, 0.028]	1.000
C-PR minus AWP	new transfer senders	0.262	0.346	-0.085	[-0.092, -0.078]	0.000
C-PR minus AWP	new approval pairs	0.252	0.196	+0.056	[0.042, 0.070]	1.000

spenders ranks lower than its count alone would place it; on this label that division costs a little. None of these degree contrasts was named in the plan; all were computed after the labels were known.

Read across the table, each label is ranked best by scores that use its own edge type. On new approval pairs the C-PR mixtures, the in-approve degree and EndorseRank lead, while AWP (0.196 $[0.181, 0.210]$), S-PR (0.172 $[0.157, 0.185]$), C-PR ($\lambda = 0$) (0.160 $[0.144, 0.174]$) and the transfer in-degree (0.135 $[0.121, 0.150]$) stay lower. On new transfer senders AWP and the transfer in-degree lead, and the allowance-side scores stay lower and close together: EndorseRank reaches 0.200 $[0.189, 0.210]$, C-PR ($\lambda = 1$) 0.188 $[0.177, 0.199]$ and the in-approve degree 0.180 $[0.169, 0.192]$, with overlapping intervals and no paired contrast among them. EndorseRank leaves almost no ties here: it gives 16,208 distinct scores to the 17,050 spenders, its largest tie block holds 156 of them, and only 0.027% of pairs are tied, so its coefficients rest on an ordering of nearly every spender.

Coverage does not explain the degree comparisons. Of the 17,050 spenders, 2,891 have no transfer edge up to the freeze and score zero under AWP. They gained new approval pairs slightly more often than the rest (13.3% against 11.9%) and new senders almost never (0.6% against 11.9%). Table 4.12 restricts the holdout to the 14,094 spenders that had received a transfer from another address before the freeze; the restriction was chosen after the results were known. There AWP still ranks new senders above the transfer in-degree ($+0.056$, $[0.049, 0.064]$), and EndorseRank still ranks new approval pairs below the in-approve degree (-0.013, $[-0.020, -0.006]$). AWP's coefficient on new approval pairs rises from 0.196 $[0.181, 0.210]$ to 0.298 $[0.287, 0.308]$ on this subset, since its zeros had placed at the bottom many spenders that later gained approvals; there its point estimate lies above those of the in-approve degree (0.249 $[0.237, 0.264]$) and EndorseRank (0.236 $[0.224, 0.249]$), although no paired contrast was computed for these pairs.

The second finding is that the coupled operator fails rule A on the revised scores, although in an exploratory comparison it improves on its transfer layer walked alone. Rule A, fixed in the analysis plan before any score of the study existed, asked C-PR at $\lambda = 0.5$ to do no worse than C-PR ($\lambda = 1$) on new approval pairs and better than C-PR ($\lambda = 0$) on new transfer senders. Here it is applied to

Table 4.12: Spender holdout W0 restricted to the 14,094 of the 17,050 spender contracts that had received a transfer from another address before the freeze ($n = 14{,}094$). The other 2,956 have no inbound transfer edge from another address, and 2,891 spenders score zero under AWP. The restriction was chosen after the results were known. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Score at t_1	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
EndorseRank	0.236 [0.224, 0.249]	0.231 [0.219, 0.244]
AWP	0.298 [0.287, 0.308]	0.357 [0.347, 0.367]
In-approve degree at t_1	0.249 [0.237, 0.264]	0.205 [0.193, 0.218]
Transfer in-degree at t_1	0.211 [0.196, 0.224]	0.301 [0.289, 0.313]
EndorseRank minus t_1 in-approve degree (new approval pairs)	$\Delta\tau = -0.013$ [-0.020, -0.006]	
AWP minus t_1 in-degree (new transfer senders)	$\Delta\tau = +0.056$ [0.049, 0.064]	

scores defined after the registered results were known, so its verdict is a re-analysis and not a registered test. The first condition holds: C-PR lies above the allowance walk ($+0.029$, $[0.022, 0.036]$). The second fails: C-PR lies below the transfer walk on new senders (-0.034, $[-0.040, -0.029]$). The registered analysis reached the same verdict (Table 8.13). With the primary part failed, the second half of RQ5, whether C-PR beats a walk over either layer alone, receives a negative answer, as the plan specified. Against the transfer walk alone on new approval pairs, a comparison that rule A did not include, C-PR does better: it reaches 0.252 $[0.241, 0.262]$ against 0.160 $[0.144, 0.174]$, a paired difference of $+0.092$ $[0.080, 0.105]$ on the same resamples. In the registered analysis the same contrast, seen after the labels were known, became the hypothesis F1 of the registration of W1; Section 4.6.2 applies F1 to the revised scores. Against EndorseRank and AWP, also computed after the labels were known, C-PR lies above EndorseRank on new approval pairs ($+0.021$, $[0.013, 0.028]$) and above AWP ($+0.056$, $[0.042, 0.070]$), and it falls well behind AWP on new senders (-0.085, $[-0.092, -0.078]$). Across the three values of λ, C-PR is flat on new approval pairs (0.253 $[0.243, 0.264]$ at $\lambda = 0.25$, 0.249 $[0.238, 0.260]$ at $\lambda = 0.75$), and its point estimate on new transfer senders falls as λ rises, from 0.269 $[0.257, 0.280]$ to 0.253 $[0.242, 0.265]$. No paired contrast between values of λ was computed, and λ was not tuned.

The seeded ablation shows where the allowance signal has to sit. S-PR is level with the transfer walk on new senders (-0.001, $[-0.004, 0.002]$) and falls below the allowance walk on new approval

pairs (-0.051, $[-0.066, -0.038]$). Its restarts lift its coefficient on new approval pairs only a little above that of the transfer walk (0.172 $[0.157, 0.185]$ against 0.160 $[0.144, 0.174]$; no paired contrast), so when transfer edges decide where the walk goes, little of the allowance signal survives. On the spender holdout, allowance edges help when they carry the walk, fully in the allowance walk and mixed in C-PR, and help little when they only decide where the walk restarts. The S-PR contrasts and the increment of C-PR ($\lambda = 1$) over its degree were computed after the labels were known; only the two contrasts of rule A were named in the plan.

Two caveats apply to the whole table. Each label is the same construction carried into the future (new approvals, new senders), so none of these spender results bears on credit or loss; the trader labels below are built from neither edge type but are not credit labels either. Both labels are also sparse. Of the 17,050 spenders, 2,070 (12.1%) gained a new approval pair and 1,695 (9.9%) a new sender, and every coefficient rests largely on how these subsets are ordered.

Two other candidate labels, which the plan did not name, were computed on the holdout, and they are reported here without being offered as support. Revocation counts rise with every score, most with the raw in-approve degree (0.388 $[0.376, 0.398]$), then with EndorseRank (0.337 $[0.327, 0.347]$), and least with the walks over transfers, AWP (0.191 $[0.175, 0.205]$) and C-PR ($\lambda = 0$) (0.200 $[0.185, 0.214]$). The drain heuristic rises with every score as well, from 0.112 $[0.096, 0.126]$ for the transfer in-degree to 0.328 $[0.318, 0.338]$ for EndorseRank with activity restarts, with EndorseRank at 0.234 $[0.224, 0.245]$ and the in-approve degree at 0.214 $[0.200, 0.227]$. For an endorsement reading the sign is wrong, since a trusted spender should lose approvals less often, not more. The likelier reading is volume. A spender with many owners sees more of everything owners do, revocations and drains included, so these labels cannot separate trust from use.

4.6.1 Trader outcomes (exploratory)

The trader run of Section 3.7.1 applies the W0 freeze to the GMX V2 contract traders and is reported as exploratory; its liquidation labels were left for the comparison of Section 4.6.3. Table 4.13 gives every score on the four profit labels of the 88 contract traders with at least three closes from April

to June, 40 of which had received an approval by the freeze. With 88 traders every interval is wide.

Table 4.13: W0 trader run: Kendall τ between scores frozen at 31 March 2026 and GMX V2 labels counted from 1 April 2026 to 30 June 2026, on the $n = 88$ GMX V2 contract accounts in the freeze-date graph with at least three closes in that window, 40 of which had received an approval by the freeze. The run is exploratory; the liquidation labels of W0 were left for the comparison (Table 4.18). Profitable closes and realized gain grow with the number of closes; the two shares do not. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Score at t_1	Profitable closes	Realized gain	Profitable share	Non-loss share
In-approve degree at t_1 (baseline)	0.083 [-0.103, 0.238]	0.024 [-0.154, 0.202]	-0.146 [-0.317, 0.029]	-0.278 [-0.431, -0.119]
Transfer in-degree at t_1 (baseline)	0.008 [-0.135, 0.169]	0.008 [-0.149, 0.175]	-0.191 [-0.329, -0.041]	-0.197 [-0.350, -0.035]
EndorseRank	0.150 [-0.019, 0.295]	0.078 [-0.094, 0.226]	-0.055 [-0.196, 0.084]	-0.178 [-0.298, -0.042]
AWP	0.155 [0.005, 0.297]	0.196 [0.048, 0.351]	-0.124 [-0.265, 0.015]	-0.254 [-0.404, -0.104]
C-PR ($\lambda = 1$)	0.083 [-0.072, 0.223]	0.040 [-0.133, 0.190]	-0.074 [-0.231, 0.068]	-0.188 [-0.322, -0.047]
C-PR ($\lambda = 0$)	0.150 [0.011, 0.272]	0.288 [0.128, 0.438]	-0.101 [-0.243, 0.045]	-0.225 [-0.359, -0.083]
C-PR ($\lambda = 0.25$)	0.152 [0.015, 0.274]	0.294 [0.143, 0.424]	-0.084 [-0.233, 0.069]	-0.210 [-0.352, -0.061]
C-PR ($\lambda = 0.5$)	0.160 [0.030, 0.284]	0.283 [0.142, 0.410]	-0.084 [-0.227, 0.058]	-0.208 [-0.346, -0.061]
C-PR ($\lambda = 0.75$)	0.155 [0.025, 0.279]	0.270 [0.127, 0.398]	-0.082 [-0.221, 0.063]	-0.203 [-0.341, -0.059]
S-PR	0.166 [0.023, 0.290]	0.301 [0.147, 0.436]	-0.075 [-0.221, 0.072]	-0.221 [-0.365, -0.078]

The two count labels separate the walks over transfers from the other scores. On profitable closes and realized gain, which grow with how much a contract trades, AWP, C-PR ($\lambda = 0$), the three mixtures and S-PR reach coefficients with intervals above zero, from 0.150 $[0.011, 0.272]$ for C-PR ($\lambda = 0$) on profitable closes to 0.301 $[0.147, 0.436]$ for S-PR on realized gain, while the intervals of EndorseRank, C-PR ($\lambda = 1$) and both raw degrees contain zero. The transfer in-degree is near zero on both labels (0.008 $[-0.135, 0.169]$ and 0.008 $[-0.149, 0.175]$), so the number of senders alone does not rank them. On the two shares, which do not grow with activity, every coefficient is

negative. On the non-loss share every interval lies below zero, from -0.178 $[-0.298, -0.042]$ for EndorseRank to -0.278 $[-0.431, -0.119]$ for the in-approve degree; on the profitable share only that of the transfer in-degree does (-0.191, $[-0.329, -0.041]$). A contract that ranks high on these scores closed a smaller share of its positions without a loss, and part of that is built into the non-loss share (Section 4.6.3). None of the scores ranks how well a contract trades. In W1, rule B sets C-PR against its transfer layer on a trader label of its own, the liquidation-free close rate (F3).

4.6.2 Window W1: July to September 2026

The registration of Section 3.7.2 was written after the W0 results of the registered analysis were known and committed to the public repository as commit `0d3a057` at 16:49:32 UTC on 9 October 2026 (01:49 on 10 October in the commit's time zone, UTC+9). It was committed before the study extracted the July–September logs, as a preregistration in the sense of Nosek et al. (2018), and fixed rule B and its margin, three further results reported beside the decision (F4–F6), two sensitivity analyses and the comparison on outcomes built from neither edge type. The margin $\delta = 0.01$ is the half-width of the W0 interval of the F2 contrast in the registered analysis (0.0040; -0.030, $[-0.034, -0.026]$), rounded up to two decimals. The extraction script refuses to run without the committed registration (Appendix 8.1). Its manifest records the registration commit, the SHA-256 hashes of the registration and of the configuration, no use of the testing override, and a start one second after the commit, at 16:49:33 UTC. The July–September logs held 6,805 GMX V2 closes by contract accounts, all of which decoded. The results below are those of the revised scores, computed after the registered evaluation of W1 was known, with the same 400 paired resamples and seed 42. Rule B, F4–F6 and the two sensitivity analyses are applied to them with the registered thresholds, as a re-analysis and not as registered tests; the registered verdicts are in Appendix 8.6.10.

At the freeze on 30 June 2026, 18,152 contract spenders held a positive latest allowance of one of the five tokens; 1,616 of them gained a new approval pair from July to September and 1,533 a new sender. Seventy GMX V2 contract traders in the freeze-date graph closed at least three positions in the label window; 9 of them had at least one liquidation, and 1 had every close liquidated. With

so few liquidations the trader label varies little, and every trader interval is wide. Table 4.14 gives every score of rule B on its three labels, and Table 4.15 the contrasts with their verdicts.

Table 4.14: Window W1: Kendall τ between scores frozen at 30 June 2026 and the three labels of rule B, counted from 1 July 2026 to 30 September 2026. The spenders are the $n = 18{,}152$ contract spenders holding a positive latest allowance at the freeze; 1,616 of them gained a new approval pair and 1,533 a new sender. The traders are the $n = 70$ GMX V2 contract accounts in the freeze-date graph with at least three closes in the label window; 9 of them had at least one liquidation and 1 had only liquidations. C-PR ($\lambda = 0$) is the comparator of rule B and AWP that of its first sensitivity analysis (Table 4.15); EndorseRank is in Table 4.16. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Spenders ($n = 18{,}152$)		Traders ($n = 70$)
Score at 30 June 2026	New approval pairs	New transfer senders	Liquidation-free close rate
In-approve degree at the freeze (baseline)	0.268 [0.257, 0.279]	0.205 [0.193, 0.215]	0.394 [0.261, 0.524]
Transfer in-degree at the freeze (baseline)	0.131 [0.119, 0.143]	0.308 [0.299, 0.319]	0.310 [0.185, 0.416]
AWP	0.169 [0.157, 0.183]	0.330 [0.321, 0.340]	0.329 [0.215, 0.439]
C-PR ($\lambda = 1$)	0.228 [0.217, 0.238]	0.188 [0.178, 0.198]	0.301 [0.177, 0.425]
C-PR ($\lambda = 0$)	0.141 [0.129, 0.154]	0.291 [0.282, 0.301]	0.350 [0.236, 0.461]
C-PR ($\lambda = 0.25$)	0.251 [0.241, 0.260]	0.270 [0.260, 0.279]	0.349 [0.234, 0.459]
C-PR ($\lambda = 0.5$)	0.250 [0.240, 0.260]	0.263 [0.254, 0.273]	0.350 [0.236, 0.461]
C-PR ($\lambda = 0.75$)	0.249 [0.238, 0.259]	0.256 [0.246, 0.266]	0.347 [0.231, 0.462]
S-PR	0.155 [0.142, 0.168]	0.291 [0.281, 0.301]	0.374 [0.255, 0.498]

Rule B, applied to the revised scores, is not met. F1 holds: C-PR ranks the spenders that later gained new approval pairs at $\tau = 0.250$ $[0.240, 0.260]$ against 0.141 $[0.129, 0.154]$ for its transfer layer walked alone, a difference of $+0.110$ $[0.098, 0.122]$, so the W0 gain of $+0.092$ $[0.080, 0.105]$ recurs in W1. F2 fails. On new transfer senders C-PR reaches 0.263 $[0.254, 0.273]$ against 0.291 $[0.282, 0.301]$; the difference is -0.028 $[-0.033, -0.023]$, and the whole interval lies below the margin of -0.01, so C-PR ranks later inflow worse than the transfer walk by more than the registration allowed. F3 fails narrowly. On the liquidation-free close rate of the traders C-PR and the transfer walk both reach 0.350 $[0.236, 0.461]$, and the difference is $+0.000$ $[-0.014, 0.010]$, whose lower end lies below -0.01. The rule needed all three hypotheses, so, as the registration said for this case, the thesis makes no claim that C-PR improves on a walk over the transfers alone. The registered analysis reached the same verdict (Table 8.13).

Table 4.15: Window W1: the paired contrasts F1–F6 of rule B with their verdicts, and the two sensitivity analyses (rule B and its sensitivity analyses were registered for the raw-unit analysis; here they are applied to the revised scores). $\Delta\tau = \tau_a - \tau_b$, with C-PR ($\lambda = 0.5$) as a. Superiority holds when the interval lies above zero; non-inferiority holds when its lower end lies above $-\delta = -0.01$; F6a, the first part of rule A, holds when the interval contains zero or lies above it. F1, F2, F6a and F6b are on the spenders ($n = 18{,}152$ contracts), F3–F5 on the traders ($n = 70$), as in Table 4.14. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Contrast	Label	Test	τ_a	τ_b	$\Delta\tau$ [95% CI]	Verdict
Rule B: C-PR ($\lambda = 0.5$) against its transfer layer walked alone, C-PR ($\lambda = 0$)							
F1	C-PR vs. C-PR ($\lambda = 0$)	new approval pairs	superiority	0.250	0.141	+0.110 [0.098, 0.122]	holds
F2	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	non-inferiority, -0.01	0.263	0.291	-0.028 [-0.033, -0.023]	fails
F3	C-PR vs. C-PR ($\lambda = 0$)	liquidation-free close rate	non-inferiority, -0.01	0.350	0.350	+0.000 [-0.014, 0.010]	fails
Decision: F1, F2 and F3 must all hold							not met
Reported, not part of the decision							
F4	C-PR vs. C-PR ($\lambda = 0$)	liquidation-free close rate	superiority	0.350	0.350	+0.000 [-0.014, 0.010]	fails
F5	C-PR vs. C-PR ($\lambda = 1$)	liquidation-free close rate	superiority	0.350	0.301	+0.049 [-0.055, 0.168]	fails
F6a	C-PR vs. C-PR ($\lambda = 1$)	new approval pairs	no worse	0.250	0.228	+0.022 [0.015, 0.029]	holds
F6b	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	superiority	0.263	0.291	-0.028 [-0.033, -0.023]	fails
Sensitivity analysis 1: AWP as the comparator							
F1	C-PR vs. AWP	new approval pairs	superiority	0.250	0.169	+0.081 [0.069, 0.095]	holds
F2	C-PR vs. AWP	new transfer senders	non-inferiority, -0.01	0.263	0.330	-0.067 [-0.072, -0.061]	fails
F3	C-PR vs. AWP	liquidation-free close rate	non-inferiority, -0.01	0.350	0.329	+0.021 [-0.019, 0.060]	fails
F4	C-PR vs. AWP	liquidation-free close rate	superiority	0.350	0.329	+0.021 [-0.019, 0.060]	fails
Sensitivity analysis 2: contracts outside a graph kept as isolated nodes							
F1	C-PR vs. C-PR ($\lambda = 0$)	new approval pairs	superiority	0.250	0.141	+0.109 [0.098, 0.121]	holds
F2	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	non-inferiority, -0.01	0.263	0.291	-0.028 [-0.033, -0.023]	fails
F3	C-PR vs. C-PR ($\lambda = 0$)	liquidation-free close rate	non-inferiority, -0.01	0.350	0.350	+0.000 [-0.014, 0.010]	fails
F4	C-PR vs. C-PR ($\lambda = 0$)	liquidation-free close rate	superiority	0.350	0.350	+0.000 [-0.014, 0.010]	fails
F5	C-PR vs. C-PR ($\lambda = 1$)	liquidation-free close rate	superiority	0.350	0.322	+0.028 [-0.047, 0.094]	fails
F6a	C-PR vs. C-PR ($\lambda = 1$)	new approval pairs	no worse	0.250	0.228	+0.022 [0.015, 0.029]	holds
F6b	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	superiority	0.263	0.291	-0.028 [-0.033, -0.023]	fails

The registration warned that F3 could pass with both coefficients near zero. Here neither coefficient is near zero, and F3 fails because the paired interval reaches below the margin although the point estimates are equal. Every score ranks this label with an interval above zero: the in-approve degree at the freeze reaches 0.394 $[0.261, 0.524]$, S-PR 0.374 $[0.255, 0.498]$, the three mixtures of C-PR between 0.347 $[0.231, 0.462]$ and 0.350 $[0.236, 0.461]$, AWP 0.329 $[0.215, 0.439]$, the transfer in-degree 0.310 $[0.185, 0.416]$ and C-PR ($\lambda = 1$) 0.301 $[0.177, 0.425]$ (Table 4.14).

The results reported beside the decision follow the registered order. F4 fails: on the liquidation-free close rate C-PR is not better than the transfer walk, and the interval of the difference contains zero. F5 fails, since C-PR and the allowance walk cannot be told apart on that label ($+0.049$, $[-0.055, 0.168]$). F6, the primary part of rule A rerun on W1, fails as in W0: C-PR is no worse than the allowance walk on new approval pairs ($+0.022$, $[0.015, 0.029]$) but lies below the transfer walk on new senders (-0.028, $[-0.033, -0.023]$). As in W0, the point estimates of C-PR are flat across λ on new approval pairs and fall on new senders as λ rises (Table 4.14). S-PR ranks new approval pairs at 0.155 $[0.142, 0.168]$, below the allowance walk (0.228 $[0.217, 0.238]$), and new senders at 0.291 $[0.281, 0.301]$, level with the transfer walk (0.291 $[0.282, 0.301]$); no paired contrast of S-PR was registered for W1.

Neither sensitivity analysis changes the verdict (Table 4.15). With AWP as the comparator, F1 holds ($+0.081$, $[0.069, 0.095]$) and F2 fails clearly. AWP ranks new senders at 0.330 $[0.321, 0.340]$, and C-PR's difference to it is -0.067 $[-0.072, -0.061]$; F3 and F4 fail as well ($+0.021$, $[-0.019, 0.060]$). Against AWP, then, C-PR gives up a measurable part of the ranking of later inflow in exchange for its gain on later approvals. With the contracts outside a graph kept as isolated nodes, every verdict stays as it was, and the point estimates of F1 to F4 and F6 change by at most 0.001. F5 stays unresolved. The rule lifts the traders outside the allowance graph from zero to the score of a node without in-edges, the allowance walk's coefficient on the liquidation-free close rate rises, and C-PR's lead over it shrinks to $+0.028$ $[-0.047, 0.094]$.

Table 4.16 adds EndorseRank and its variant with activity restarts. Neither is a score of rule B, and none of their rows enters the decision. None of the contrasts in this table was named in the registra-

tion; all were computed after the July–September labels were known, those of the variant afterwards from the saved scores. The pattern of W0 holds for both single-layer walks. The in-approve degree at the freeze again ranks new approval pairs above EndorseRank (0.268 $[0.257, 0.279]$ against 0.234 $[0.222, 0.244]$; -0.034, $[-0.038, -0.029]$), and AWP again ranks new senders above the transfer in-degree (0.330 $[0.321, 0.340]$ against 0.308 $[0.299, 0.319]$; $+0.022$, $[0.017, 0.027]$). With activity restarts EndorseRank ranks new approval pairs above the degree ($+0.027$, $[0.021, 0.034]$), as it did in W0. C-PR ranks new approval pairs above EndorseRank ($+0.016$, $[0.010, 0.024]$), as it did in W0, and the two cannot be told apart on the liquidation-free close rate ($+0.053$, $[-0.051, 0.171]$).

Table 4.16: EndorseRank and its variant with activity restarts on the cohorts and labels of the window W1 (spenders $n = 18{,}152$ contracts, traders $n = 70$). Neither these rows nor the contrasts below them are part of rule B, and the contrasts were not named in the registration. The contrasts of EndorseRank with activity restarts were computed afterwards from the saved W1 scores. The AWP, C-PR and degree rows repeat Table 4.14, and every contrast uses the same contract resamples; the differences between EndorseRank and AWP are in Table 8.11. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Spenders ($n = 18{,}152$)		Traders ($n = 70$)
Score at 30 June 2026	New approval pairs	New transfer senders	Liquidation-free close rate
In-approve degree at the freeze (baseline)	0.268 [0.257, 0.279]	0.205 [0.193, 0.215]	0.394 [0.261, 0.524]
Transfer in-degree at the freeze (baseline)	0.131 [0.119, 0.143]	0.308 [0.299, 0.319]	0.310 [0.185, 0.416]
EndorseRank	0.234 [0.222, 0.244]	0.194 [0.184, 0.205]	0.297 [0.174, 0.418]
EndorseRank, restarts by approving activity	0.295 [0.285, 0.304]	0.242 [0.232, 0.253]	0.288 [0.161, 0.405]
AWP	0.169 [0.157, 0.183]	0.330 [0.321, 0.340]	0.329 [0.215, 0.439]
C-PR ($\lambda = 0.5$)	0.250 [0.240, 0.260]	0.263 [0.254, 0.273]	0.350 [0.236, 0.461]
EndorseRank minus in-approve degree, new approval pairs	$\Delta\tau = -0.034$ [-0.038, -0.029]		
AWP minus transfer in-degree, new transfer senders	$\Delta\tau = +0.022$ [0.017, 0.027]		
C-PR minus EndorseRank, new approval pairs	$\Delta\tau = +0.016$ [0.010, 0.024]		
C-PR minus EndorseRank, liquidation-free close rate	$\Delta\tau = +0.053$ [-0.051, 0.171]		
EndorseRank (activity restarts) minus in-approve degree, new approval pairs	$\Delta\tau = +0.027$ [0.021, 0.034]		
EndorseRank (activity restarts) minus EndorseRank, new approval pairs	$\Delta\tau = +0.061$ [0.054, 0.067]		
EndorseRank (activity restarts) minus EndorseRank, new transfer senders	$\Delta\tau = +0.048$ [0.042, 0.053]		

Table 4.17 gives the trader labels of W1 that do not enter the decision, with the 2,000 resamples of Section 4.6.3. The scores built on transfers rank the two count labels. On profitable closes AWP reaches 0.288 $[0.125, 0.442]$ and C-PR ($\lambda = 0$) 0.199 $[0.043, 0.355]$, and on realized gain AWP reaches 0.346 $[0.189, 0.490]$ and C-PR ($\lambda = 0$) 0.338 $[0.178, 0.485]$, while the allowance-side scores stay near zero on both. On the no-liquidation flag every score is positive, the in-

approve degree highest at 0.407 $[0.268, 0.541]$ and EndorseRank with activity restarts lowest at 0.294 $[0.158, 0.417]$. On the two shares the allowance-side scores are negative, and AWP and C-PR ($\lambda = 0$) are positive with intervals that contain zero. On the non-loss share the intervals of the in-approve degree (-0.294, $[-0.468, -0.106]$), C-PR ($\lambda = 1$) (-0.287, $[-0.443, -0.121]$), EndorseRank (-0.264, $[-0.408, -0.106]$) and its variant with activity restarts (-0.269, $[-0.412, -0.118]$) lie below zero, and on the profitable share those of the same four (-0.221, $[-0.406, -0.034]$; -0.208, $[-0.370, -0.030]$; -0.180, $[-0.340, -0.012]$; -0.199, $[-0.346, -0.039]$).

Table 4.17: Window W1, traders: Kendall τ between scores frozen at 30 June 2026 and the other GMX V2 labels counted from 1 July 2026 to 30 September 2026, on the $n = 70$ contract accounts with at least three closes in that window. No liquidation is 1 when none of the account's closes was a liquidation. Profitable closes and realized gain grow with the number of closes; the two shares do not. These labels are reported and do not enter rule B. The intervals use the resamples of Table 4.18. CI = 95% percentile bootstrap over contracts (2,000 paired resamples, seed 42).

Score at 30 June 2026	No liquidation	Profitable closes	Realized gain	Profitable share	Non-loss share
In-approve degree at the freeze (baseline)	0.407 [0.268, 0.541]	0.042 [-0.158, 0.243]	0.033 [-0.164, 0.230]	-0.221 [-0.406, -0.034]	-0.294 [-0.468, -0.106]
Transfer in-degree at the freeze (baseline)	0.329 [0.189, 0.451]	0.185 [0.014, 0.341]	0.168 [0.028, 0.291]	-0.025 [-0.194, 0.131]	-0.042 [-0.203, 0.114]
EndorseRank	0.303 [0.172, 0.422]	0.073 [-0.115, 0.260]	0.000 [-0.198, 0.200]	-0.180 [-0.340, -0.012]	-0.264 [-0.408, -0.106]
EndorseRank, restarts by approving activity	0.294 [0.158, 0.417]	0.057 [-0.123, 0.239]	-0.004 [-0.179, 0.181]	-0.199 [-0.346, -0.039]	-0.269 [-0.412, -0.118]
AWP	0.333 [0.207, 0.447]	0.288 [0.125, 0.442]	0.346 [0.189, 0.490]	0.136 [-0.039, 0.300]	0.082 [-0.077, 0.247]
C-PR ($\lambda = 1$)	0.307 [0.174, 0.428]	0.010 [-0.167, 0.192]	-0.066 [-0.249, 0.121]	-0.208 [-0.370, -0.030]	-0.287 [-0.443, -0.121]
C-PR ($\lambda = 0$)	0.358 [0.228, 0.475]	0.199 [0.043, 0.355]	0.338 [0.178, 0.485]	0.136 [-0.034, 0.290]	0.038 [-0.127, 0.201]
C-PR ($\lambda = 0.25$)	0.357 [0.227, 0.474]	0.155 [0.002, 0.305]	0.284 [0.122, 0.435]	0.090 [-0.075, 0.242]	-0.011 [-0.173, 0.148]
C-PR ($\lambda = 0.5$)	0.358 [0.229, 0.475]	0.114 [-0.044, 0.267]	0.240 [0.075, 0.397]	0.053 [-0.115, 0.212]	-0.053 [-0.217, 0.103]
C-PR ($\lambda = 0.75$)	0.353 [0.226, 0.472]	0.081 [-0.078, 0.244]	0.201 [0.023, 0.368]	0.016 [-0.153, 0.182]	-0.090 [-0.256, 0.072]
S-PR	0.381 [0.248, 0.495]	0.167 [0.018, 0.312]	0.286 [0.122, 0.441]	0.100 [-0.048, 0.244]	-0.011 [-0.164, 0.151]

4.6.3 Outcomes built from neither edge type

The GMX trader labels are the only outcomes in these data built from neither edge type. The comparison of Section 3.7.3 was registered with rule B, before the study extracted the July–September logs and before it had computed the liquidation labels of W0 for any score. Here its contrasts, rules and wording are applied to the revised scores, after the registered results were known; the verdicts below are a re-analysis, and the registered analysis reached the same ones (Appendix 8.6.10). W1 is the primary window and W0 the secondary one. Intervals come from 2,000 paired resamples and are 95% unless marked 97.5%, the Bonferroni interval of the decision contrasts P and Q. Table 4.18 gives the contrasts on the liquidation-free close rate, Table 4.19 the traders that receive approvals, and Table 8.12 (Appendix 8.5.4) all six trader labels. The samples are small, 70 contract traders in W1 and 88 in W0, and every interval below is wide.

R1 does not hold. In W1 the in-approve degree at the freeze reaches $\tau = 0.394$ $[0.265, 0.518]$ on the liquidation-free close rate and the transfer in-degree 0.310 $[0.184, 0.425]$, so P, their difference, is $+0.084$ $[-0.013, 0.176]$, with a 97.5% interval of $[-0.026, 0.188]$ that includes zero. Stratified by the number of closes, P is $+0.101$ $[-0.024, 0.231]$. In W0, P is $+0.182$ (97.5% interval $[0.032, 0.338]$), positive as R1 required, but the rule also needed both W1 intervals above zero. In the registered wording, there is no count-level lead of allowance over transfer on later liquidation avoidance. R2 does not hold either. Q, EndorseRank with AWP's activity restarts minus AWP (the like-for-like PageRank pair), is -0.041 in W1 (97.5% interval $[-0.148, 0.064]$) and -0.014 in W0 (97.5% interval $[-0.161, 0.151]$), so, in the registered wording, the PageRank versions are not resolved. Neither are the other pairs. C-PR's two layers differ by -0.049 $[-0.169, 0.060]$ in W1 and -0.041 $[-0.187, 0.105]$ in W0, EndorseRank and AWP by -0.032 $[-0.133, 0.066]$ and -0.017 $[-0.156, 0.128]$, and C-PR ($\lambda = 0$) and AWP by $+0.021$ $[-0.016, 0.065]$ and $+0.025$ $[-0.022, 0.077]$. In a contrast that the registration did not name and that was computed after the labels were known, EndorseRank falls below the approval count in both windows, by -0.120 $[-0.203, -0.033]$ in W0 and -0.097 $[-0.174, -0.009]$ in W1.

Table 4.18: Allowance-side against transfer-side scores on the liquidation-free close rate of the GMX V2 contract traders with at least three closes in the label window, a label built from neither edge type ($n = 70$ traders in W1, $n = 88$ in W0). The pairs, the decision contrasts P and Q with their Bonferroni 97.5% intervals and the stratified rows were registered before the study extracted the July–September logs, for the raw-unit analysis, and are applied here to the revised scores; W1 is the primary window and W0 the secondary one. Paired contract bootstrap with 2,000 resamples (seed 42). The stratified rows combine Kendall's τ_b within strata of the number of label-window closes (3–4, 5–9, 10–24, 25 or more), weighted by the number of contract pairs. Registered rules applied to the revised scores: R1 (count-level lead) not met; R2 (PageRank-level lead) not met; R3 (specificity) applies.

Contrast (a minus b)	τ_a	τ_b	$\Delta\tau$	95% CI	97.5% CI
Window W1, July–September 2026 ($n = 70$ traders)					
In-approve degree minus transfer in-degree	0.394	0.310	0.084	[-0.013, 0.176]	[-0.026, 0.188]
EndorseRank (activity restarts) minus AWP	0.288	0.329	-0.041	[-0.135, 0.050]	[-0.148, 0.064]
C-PR ($\lambda = 1$) minus C-PR ($\lambda = 0$)	0.301	0.350	-0.049	[-0.169, 0.060]	–
EndorseRank minus AWP	0.297	0.329	-0.032	[-0.133, 0.066]	–
C-PR ($\lambda = 0$) minus AWP	0.350	0.329	0.021	[-0.016, 0.065]	–
EndorseRank minus in-approve degree	0.297	0.394	-0.097	[-0.174, -0.009]	–
In-approve degree minus transfer in-degree, stratified by closes	0.381	0.280	0.101	[-0.024, 0.231]	–
EndorseRank (activity restarts) minus AWP, stratified by closes	0.273	0.301	-0.028	[-0.207, 0.098]	–
Window W0, April–June 2026 ($n = 88$ traders)					
In-approve degree minus transfer in-degree	0.344	0.161	0.182	[0.052, 0.314]	[0.032, 0.338]
EndorseRank (activity restarts) minus AWP	0.227	0.240	-0.014	[-0.147, 0.130]	[-0.161, 0.151]
C-PR ($\lambda = 1$) minus C-PR ($\lambda = 0$)	0.225	0.265	-0.041	[-0.187, 0.105]	–
EndorseRank minus AWP	0.224	0.240	-0.017	[-0.156, 0.128]	–
C-PR ($\lambda = 0$) minus AWP	0.265	0.240	0.025	[-0.022, 0.077]	–
EndorseRank minus in-approve degree	0.224	0.344	-0.120	[-0.203, -0.033]	–
In-approve degree minus transfer in-degree, stratified by closes	0.338	0.209	0.129	[0.020, 0.293]	–
EndorseRank (activity restarts) minus AWP, stratified by closes	0.215	0.252	-0.037	[-0.173, 0.125]	–

R3 applies. In W1, P lies below zero on the profitable share (-0.196, $[-0.364, -0.046]$) and on the non-loss share (-0.252, $[-0.391, -0.120]$), so, in the registered wording, the signal concerns forced-liquidation avoidance, not trading success. Part of the non-loss gap is built in. The pipeline counts a liquidated close as a close that was not a loss, so a trader with liquidations gains non-loss share, whereas the profitable share counts a liquidated close as not profitable; R3 therefore rests mainly on the profitable share. The other labels do not enter the decision (Table 8.12). On the no-liquidation flag, P is $+0.078$ $[-0.012, 0.166]$ in W1 and $+0.197$ $[0.059, 0.333]$ in W0. On profitable closes and realized gain, which grow with activity, the PageRank pairs favor the transfer side in W1, with intervals below zero for Q (-0.231, $[-0.429, -0.027]$, and -0.350, $[-0.536, -0.157]$), for EndorseRank against AWP (-0.214, $[-0.423, -0.012]$, and -0.346, $[-0.547, -0.144]$) and for C-PR's layers on realized gain (-0.404, $[-0.624, -0.182]$); in W1 the same three pairs also favor the transfer side on both shares (Table 8.12). P points the same way on the two count labels without being resolved, and in W0, among the pairs of an allowance-side and a transfer-side score, only C-PR's layers on realized gain are resolved (-0.248, $[-0.433, -0.067]$).

The count-level comparison rests on few traders. In W1 the highest in-approve degree among the traders is 1, so P in effect compares the 37 recipients with the 33 other traders; in W0 the maximum is 3, and 40 of the 88 traders are recipients. Every recipient avoided liquidation in both windows (37 of 37 and 40 of 40), against 72.7% of the other traders in W1 and 75.0% in W0. Recipients also closed more positions (median 24 against 11 in W1, 31 against 10 in W0) and had lower profitable and non-loss shares (mean profitable share 0.325 against 0.479 in W1), the pattern behind R3. Without the ten recipients of highest in-approve degree, which in W1 are taken from tied values in node order, P is $+0.073$ $[-0.039, 0.186]$ in W1 and $+0.188$ $[0.046, 0.331]$ in W0. Every trader in both windows holds contract code, so account type does not separate the recipients from the others. The transfer count also ranks the liquidation-free close rate in W1, and with 70 and 88 traders the lead of the approval count over it is not resolved in the window that the registration made primary.

Table 4.19: GMX V2 contract traders with a positive in-approve degree at the freeze (recipients) against all other labelled traders, in both windows ($n = 70$ traders in W1, $n = 88$ in W0). The comparison of recipients with the other traders and the influence check were registered with the contrasts of Table 4.18 and are applied here to the revised scores. The highest in-approve degree among the traders is 1 in W1 and 3 in W0, so the ten recipients left out by the influence check are taken from tied values in node order. The influence check is a paired contract bootstrap on the remaining traders (2,000 resamples, seed 42).

	Recipients	Other traders
Window W1, July–September 2026 ($n = 70$ traders)		
Traders	37	33
Share with no liquidation	1.000	0.727
Mean liquidation-free close rate	1.000	0.878
Mean profitable share	0.325	0.479
Mean non-loss share	0.433	0.644
Median closes in the label window	24.0	11.0
P without the ten recipients of highest in-approve degree: $\Delta\tau = 0.073$ [-0.039, 0.186]		
Window W0, April–June 2026 ($n = 88$ traders)		
Traders	40	48
Share with no liquidation	1.000	0.750
Mean liquidation-free close rate	1.000	0.919
Mean profitable share	0.362	0.474
Mean non-loss share	0.402	0.607
Median closes in the label window	31.0	10.0
P without the ten recipients of highest in-approve degree: $\Delta\tau = 0.188$ [0.046, 0.331]		

4.7 Claims (C1–C6)

The chapter's results support the claims below, cross-referenced to the hypothesis and research question columns of Table 1.2. All of them rest on the revised analysis. The verdicts of rules A and B in C5 and of R1–R3 in C6 apply the registered rules to the revised scores, after the registered results were known, so they are a re-analysis; the registered analysis reached the same verdicts (Appendix 8.6.10).

C1. Cost. On the full graphs of the same contracts the solver's time follows the edge count of its graph, and on the subsamples it does so from the stage of 10,000 contracts on. The allowance graph has 44.8% of the transfer graph's edges (11,210,114 against 25,001,216 at T_{obs}), and EndorseRank's call takes correspondingly less time than AWP's: AWP takes 2.17 times as long (22.37 s on 7,882,356 nodes against 10.29 s on 6,219,063 nodes), at an edge ratio of 2.23 (Table 4.1). C-PR (36.16 s on the union of 9,151,010 nodes and 31,552,359 edges) and S-PR (34.42 s on 7,884,602 nodes and 25,018,892 edges) walk the transfer layer and take longer than AWP. The speed belongs to the sparser graph, not to the algorithm.

C2. Same-window agreement. Each score agrees most with the proxies of the edge type it walks on, EndorseRank with the allowance family (0.585, $[0.578, 0.592]$) and AWP with the transfer family (0.572, $[0.564, 0.580]$), with intervals far from zero (Table 4.3). This is an intended-construct check, and the agreement is partly built in. With US-dollar amounts AWP follows the summed value of the inbound transfers almost as closely as their number of senders (in-value 0.566, $[0.558, 0.575]$; in-degree 0.578, $[0.569, 0.587]$). Every contract in the cohort received approvals of the five tokens, so EndorseRank leaves few ties (13,625 distinct scores, 0.17% of pairs tied), and the isolated-node rule changes no coefficient (Table 4.5). The C-PR mixtures lie between their two layers on every family.

C3. Divergence. EndorseRank and AWP order the contracts differently, with $\tau_b = 0.174$ $[0.163, 0.185]$; 96.7% of the pairs are ordered by both scores, and 41.2% of those are discordant (Table 4.5, Figure 4.3).

C4. Degree baselines out of window. The two walks behave differently against the counts they smooth. AWP ranks later new transfer senders better than the raw transfer in-degree in both windows (W0 $+0.043$, $[0.038, 0.048]$; W1 $+0.022$, $[0.017, 0.027]$), so on the transfer side the walk, with its time and value weights and its activity restarts, added information to the count. EndorseRank, with the uniform restarts fixed in the plan, ranks new approval pairs below the raw in-approve degree in both windows (W0 -0.019, $[-0.026, -0.011]$; W1 -0.034, $[-0.038, -0.029]$). The variant with AWP's activity restarts ranks them above the degree in both windows (W0 $+0.059$, $[0.050, 0.067]$; W1 $+0.027$, $[0.021, 0.034]$), but that comparison was made after the labels were known and is a hypothesis, not a finding (Section 4.5.4, Table 4.16). All degree contrasts were computed after the labels were known; none was named in the plan or the registration. The increment of C-PR ($\lambda = 1$) over the in-approve degree in W0 (-0.027, $[-0.034, -0.019]$) and the two S-PR contrasts were also computed after the labels. Restricting W0 to the 14,094 spenders that had received a transfer leaves both degree comparisons in place (EndorseRank -0.013, $[-0.020, -0.006]$; AWP $+0.056$, $[0.049, 0.064]$; Table 4.12).

C5. Coupling. On the revised scores, one walk over both edge types does not beat its layers walked alone. Rule A fails: C-PR is no worse than the allowance walk on new approval pairs ($+0.029$, $[0.022, 0.036]$) but worse than the transfer walk on new senders (-0.034, $[-0.040, -0.029]$), and its same-window part fails as well (Tables 4.11 and 4.4). Rule B is not met (Table 4.15). F1 holds ($+0.110$, $[0.098, 0.122]$, over the transfer walk on new approval pairs), so the W0 gain of $+0.092$ ($[0.080, 0.105]$), computed after the labels were known, recurs; F2 fails (-0.028, $[-0.033, -0.023]$, beyond the margin of 0.01); and F3 fails narrowly ($+0.000$, $[-0.014, 0.010]$, on the traders' liquidation-free close rate, with equal point estimates of 0.350 but a lower end below -0.01). With AWP as the comparator, in the first sensitivity analysis, F2 fails clearly (-0.067, $[-0.072, -0.061]$). In two W0 contrasts computed after the labels were known, S-PR is level with the transfer walk on new senders (-0.001, $[-0.004, 0.002]$) and below the allowance walk on new approval pairs (-0.051,

$[-0.066, -0.038]$): allowance edges help only when they carry the walk. Section 4.3.1 reports where the hybrids stand within the window.

C6. Outcomes built from neither edge type. On the liquidation-free close rate of GMX V2 contract traders, the count-level lead of the approval count over the transfer count (R1) is not supported in the primary window W1 (P $= +0.084$, Bonferroni 97.5% interval $[-0.026, 0.188]$; stratified $+0.101$, $[-0.024, 0.231]$), although P is positive in W0 ($+0.182$, 97.5% interval $[0.032, 0.338]$). The PageRank lead (R2) is not supported (Q $= -0.041$, 97.5% interval $[-0.148, 0.064]$; in W0 -0.014, 97.5% interval $[-0.161, 0.151]$). R3 applies: the approval count ranks traders lower than the transfer count on the profitable share (-0.196, $[-0.364, -0.046]$) and the non-loss share (-0.252, $[-0.391, -0.120]$). Every trader that had received an approval avoided liquidation in both windows (37 of 37 and 40 of 40), against 72.7% and 75.0% of the others, but the samples are small (70 and 88 contract traders), and the transfer count also ranks this label (0.310, $[0.184, 0.425]$, in W1) (Tables 4.18 and 4.19).

The rank divergence answers RQ1, the same-window tables RQ2, the benchmark tables RQ3, and the holdouts RQ4 and RQ5; C6 adds no research question. The direct comparisons between EndorseRank and AWP are in Appendix 8.5 and Table 4.18, and none of them carries a claim of the thesis. Chapter 5 reads these findings against the main research question, what ERC-20 allowance edges add to PageRank as a reputation score for smart contracts on Arbitrum One compared with the transfer walk and with the raw degrees that both walks smooth, and against the objectives of Chapter 1.